

Equipamento em Teste - Ajustes do dispositivo

Subestação/Vão:

Subestação: Asa Branca
Bay: LT SAB

Endereço da Subestação: Substation address
Endereço do Vão: bay address

Dispositivo:

Nome/descrição: SEL 311L
Tipo de Dispositivo: Line Current Differential Protection
Número Modelo/Série: serial no.
Info adicional 1: UPPL1
Info adicional 2: feeder address

Fabricante: SEL
Endereço do dispositivo: device address

Valores Nominais:

f nom: 60,00 Hz
V nom (secundária): 110,0 V
I nom (secundária): 5,000 A

Número de fases: 3
V primária: 132,0 kV
I primária: 1,200 kA

Fator para corrente/tensão residual:

VLN / VN: 1,732
VN (secundário): 36,66 V
Direção da tensão residual: 3 * V0

IN / I nom: 1,000
IN (secundário): 1,000 A
Direção da corrente residual: -3 * I0

Limites:

V max: 500,0 V

I max: 250,0 A

Filtros Debounce/Deglitch:

T de Debounce: 3,000 ms

T de Deglitch: 0,00 s

Detecção de Sobrecarga:

Tempo de Supressão: 50,00 ms

Outras propriedades do equipamento:

Tempo de queda: 20,00 ms

Objeto sob teste - Ajustes do usuário

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ General Settings

Descrição	Nome	Valor
Relay Identifier Lables	Relay Identifier Lables	
Relay Identifier (30 chars)	RID Relay Identifier (30 chars)	UPPL1_LT JOAO CAMARA III
Terminal Identifier (30 chars)	TID Terminal Identifier (30 chars)	UEE ASA BRANCA
Current Transformer Ratio and Application Settings	Current Transformer Ratio and Application Settings	
Local Phase (IA,IB,IC) CT Ratio, CTR:1	CTR Local Phase (IA,IB,IC) CT Ratio, CTR:1	240
Application	APP Application	311L
Advance Setting Enable	EADVS Advance Setting Enable	Y

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Line Current Differential Settings

Descrição	Nome	Valor
Line Current Differential Configuration Settings	Line Current Differential Configuration Settings	
Number of 87L Terminals	E87L Number of 87L Terminals	2
Enable High Speed Tripping	EHST Enable High Speed Tripping	N
Enable High Speed Direct Transfer Trip	EHSDDT Enable High Speed Direct Transfer Trip	Y
Enable Delta Current Detect	EDD Enable Delta Current Detect	Y
ETAP Tapped Load Coordination	ETAP Tapped Load Coordination	N
Enable Open CT Logic	EOCTL Enable Open CT Logic	N
Primary 87L Channel	PCHAN Primary 87L Channel	X
Hot-Standby Channel Feature	EHSC Hot-Standby Channel Feature	N
CTR at Terminal Connected to Channel X	CTR_X CTR at Terminal Connected to Channel X	240

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Line Current Differential Settings \ 87L Settings

Descrição	Nome	Valor
Minimum Difference Current Enable Level Settings	Minimum Difference Current Enable Level Settings	
Phase 87L (Amps secondary)	87LPP Phase 87L (Amps secondary)	6,00 A
Negative-Sequence 87L (Amps secondary)	87L2P Negative-Sequence 87L (Amps secondary)	0,50 A
Ground 87L (Amps secondary)	87LGP Ground 87L (Amps secondary)	+inf A
Ph. Diff. Current Alarm Pickup (Amps secondary)	CTALARM Ph. Diff. Current Alarm Pickup (Amps secondary)	0,50 A
Restraint Region Characteristic Settings	Restraint Region Characteristic Settings	
Outer Radius	87LR Outer Radius	6,0
Angle (degrees)	87LANG Angle (degrees)	195 °

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Backup Protection and Line Parameters

Descrição	Nome	Valor
Backup Protection Transformer Ratio Settings	Backup Protection Transformer Ratio Settings	
Polarizing (IPOL) CT Ratio, CTRP:1	CTRP Polarizing (IPOL) CT Ratio, CTRP:1	200
Phase (VA,VB,VC) PT Ratio, PTR:1	PTR Phase (VA,VB,VC) PT Ratio, PTR:1	1200,00
Synch. Voltage (VS) PT Ratio, PTRS:1	PTRS Synch. Voltage (VS) PT Ratio, PTRS:1	1200,00
Line Settings	Line Settings	
Pos-Seq Line Impedance Magnitude (Ohms secondary)	Z1MAG Pos-Seq Line Impedance Magnitude (Ohms secondary)	1,23 Ohm
Pos-Seq Line Impedance Angle (degrees)	Z1ANG Pos-Seq Line Impedance Angle (degrees)	79,39 °
Zero-Seq Line Impedance Magnitude (Ohms secondary)	Z0MAG Zero-Seq Line Impedance Magnitude (Ohms secondary)	4,89 Ohm
Zero-Seq Line Impedance Angle (degrees)	Z0ANG Zero-Seq Line Impedance Angle (degrees)	78,68 °
Line Length (unitless)	LL Line Length (unitless)	13,10
Fault Locator	Fault Locator	
Fault Locator Enable	EFLOC Fault Locator Enable	Y

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Phase Distance

Descrição	Nome	Valor
Enable Mho Phase Distance Elements	E21P Enable Mho Phase Distance Elements	4
CCVT Transient Detection Enable	ECCVT CCVT Transient Detection Enable	N
Mho Phase Distance Element Reach Settings	Mho Phase Distance Element Reach Settings	
Reach Zone1 (Ohm secondary)	Z1P Reach Zone1 (Ohm secondary)	0,98 Ohm
Reach Zone2 (Ohm secondary)	Z2P Reach Zone2 (Ohm secondary)	1,59 Ohm
Reach Zone3 (Ohm secondary)	Z3P Reach Zone3 (Ohm secondary)	1,47 Ohm
Reach Zone4 (Ohm secondary)	Z4P Reach Zone4 (Ohm secondary)	1,96 Ohm
Mho Phase Distance Overcurrent Fault Detector Settings	Mho Phase Distance Overcurrent Fault Detector Settings	
Phase-Phase Overcurrent Fault Detector Zone1 (Amps secondary)	50PP1 Phase-Phase Overcurrent Fault Detector Zone1 (Amps secondary)	0,50 A
Phase-Phase Overcurrent Fault Detector Zone2 (Amps secondary)	50PP2 Phase-Phase Overcurrent Fault Detector Zone2 (Amps secondary)	0,50 A
Phase-Phase Overcurrent Fault Detector Zone3 (Amps secondary)	50PP3 Phase-Phase Overcurrent Fault Detector Zone3 (Amps secondary)	0,50 A
Phase-Phase Overcurrent Fault Detector Zone4 (Amps secondary)	50PP4 Phase-Phase Overcurrent Fault Detector Zone4 (Amps secondary)	0,50 A

Descrição	Nome	Valor
Enable Mho Ground Distance Elements	E21MG Enable Mho Ground Distance Elements	N
E21XG Enable Quad Ground Distance Elements	E21XG Enable Quad Ground Distance Elements	
Enable Quad Ground Distance Elements	E21XG Enable Quad Ground Distance Elements	4
Reactance Zone1 Quad Ground (Ohm secondary)	XG1 Quad Zone1 Reactance (Ohm secondary)	0,98 Ohm
Reactance Zone2 Quad Ground (Ohm secondary)	XG2 Quad Zone2 Reactance (Ohm secondary)	1,59 Ohm
Reactance Zone3 Quad Ground (Ohm secondary)	XG3 Quad Zone3 Reactance (Ohm secondary)	1,47 Ohm
Reactance Zone4 Quad Ground (Ohm secondary)	XG4 Quad Zone4 Reactance (Ohm secondary)	1,96 Ohm
Resistance Zone1 Quad Ground (Ohm secondary)	RG1 Quad Zone1 Resistance (Ohm secondary)	4,00 Ohm
Resistance Zone2 Quad Ground (Ohm secondary)	RG2 Quad Zone2 Resistance (Ohm secondary)	12,00 Ohm
Resistance Zone3 Quad Ground (Ohm secondary)	RG3 Quad Zone3 Resistance (Ohm secondary)	12,00 Ohm
Resistance Zone4 Quad Ground (Ohm secondary)	RG4 Quad Zone4 Resistance (Ohm secondary)	12,00 Ohm
Quad Ground Polarizing Quantity	XGPOL Quad Ground Polarizing Quantity	I2
None Homogeneous Correction Angle (degrees)	TANG None Homogeneous Correction Angle (degrees)	-2,00 °
Quad and Mho Ground Distance Fault Detector Settings	Quad and Mho Ground Distance Fault Detector Settings	
Phase Overcurrent Fault Detector Zone1 for Ground Fault (Amps secondary)	50L1 Zone1 Phase Current FD (Amps secondary)	0,50 A
Phase Overcurrent Fault Detector Zone2 for Ground Fault (Amps secondary)	50L2 Zone2 Phase Current FD (Amps secondary)	0,50 A
Phase Overcurrent Fault Detector Zone3 for Ground Fault (Amps secondary)	50L3 Zone3 Phase Current FD (Amps secondary)	0,50 A
Phase Overcurrent Fault Detector Zone4 for Ground Fault (Amps secondary)	50L4 Zone4 Phase Current FD (Amps secondary)	0,50 A
Residual Current Fault Detector Zone1 for Ground Fault (Amps secondary)	50GZ1 Zone1 Residual Current FD (Amps secondary)	0,50 A
Residual Current Fault Detector Zone2 for Ground Fault (Amps secondary)	50GZ2 Zone2 Residual Current FD (Amps secondary)	0,50 A
Residual Current Fault Detector Zone3 for Ground Fault (Amps secondary)	50GZ3 Zone3 Residual Current FD (Amps secondary)	0,50 A
Residual Current Fault Detector Zone4 for Ground Fault (Amps secondary)	50GZ4 Zone4 Residual Current FD (Amps secondary)	0,50 A
Zero-Sequence Compensation Settings	Zero-Sequence Compensation Settings	
Zone1 ZSC Factor Magnitude	K0M1 Zone1 ZSC Factor Magnitude	0,995
Zone1 ZSC Factor Angle	K0A1 Zone1 ZSC Factor Angle	-0,95 °
Zone2,3,4 ZSC Factor Magnitude	K0M Zone2,3,4 ZSC Factor Magnitude	0,995
Zone2,3,4 ZSC Factor Angle	K0A Zone2,3,4 ZSC Factor Angle	-0,95 °
Mho Phase Distance Element Time Delay Settings	Mho Phase Distance Element Time Delay Settings	
Zone1 Phase Time Delay (cycles in 0.25 increments)	Z1PD Zone1 Phase Time Delay (cycles in 0.25 increments)	0,00 cy
Zone2 Phase Time Delay (cycles in 0.25 increments)	Z2PD Zone2 Phase Time Delay (cycles in 0.25 increments)	24,00 cy
Zone3 Phase Time Delay (cycles in 0.25 increments)	Z3PD Zone3 Phase Time Delay (cycles in 0.25 increments)	90,00 cy
Zone4 Phase Time Delay (cycles in 0.25 increments)	Z4PD Zone4 Phase Time Delay (cycles in 0.25 increments)	90,00 cy
Mho Ground Distance Element Time Delay Settings	Mho Ground Distance Element Time Delay Settings	

Zone1 Ground Time Delay (cycles in 0.25 increments)	Z1GD Zone1 Ground Time Delay (cycles in 0.25 increments)	0,00 cy
Zone2 Ground Time Delay (cycles in 0.25 increments)	Z2GD Zone2 Ground Time Delay (cycles in 0.25 increments)	24,00 cy
Zone3 Ground Time Delay (cycles in 0.25 increments)	Z3GD Zone3 Ground Time Delay (cycles in 0.25 increments)	90,00 cy
Zone4 Ground Time Delay (cycles in 0.25 increments)	Z4GD Zone4 Ground Time Delay (cycles in 0.25 increments)	90,00 cy
Common Phase/Ground Dist. Element Time Delay Settings	Common Phase/Ground Dist. Element Time Delay Settings	
Zone1 Common Ph/Gr Time Delay (cycles in 0.25 increments)	Z1D Zone1 Common Ph/Gr Time Delay (cycles in 0.25 increments)	0,00 cy
Zone2 Common Ph/Gr Time Delay (cycles in 0.25 increments)	Z2D Zone2 Common Ph/Gr Time Delay (cycles in 0.25 increments)	24,00 cy
Zone3 Common Ph/Gr Time Delay (cycles in 0.25 increments)	Z3D Zone3 Common Ph/Gr Time Delay (cycles in 0.25 increments)	90,00 cy
Zone4 Common Ph/Gr Time Delay (cycles in 0.25 increments)	Z4D Zone4 Common Ph/Gr Time Delay (cycles in 0.25 increments)	90,00 cy

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Phase Instantaneous Overcurrent

Descrição	Nome	Valor
Enable Phase Overcurrent Elements	E50P Enable Phase Overcurrent Elements	3
Level 1 (Amps secondary)	50P1P Level 1 (Amps secondary)	+inf A
Level 2 (Amps secondary)	50P2P Level 2 (Amps secondary)	5,00 A
Level 3 (Amps secondary)	50P3P Level 3 (Amps secondary)	28,00 A
Phase Definite-Time Overcurrent Elements	Phase Definite-Time Overcurrent Elements	
Level 1 (cycles in 0.25 increments)	67P1D Level 1 (cycles in 0.25 increments)	0,00 cy
Level 2 (cycles in 0.25 increments)	67P2D Level 2 (cycles in 0.25 increments)	90,00 cy
Level 3 (cycles in 0.25 increments)	67P3D Level 3 (cycles in 0.25 increments)	0,00 cy

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Residual Ground Instantaneous Overcurrent

Descrição	Nome	Valor
Enable Residual Ground Overcurrent Elements	E50G Enable Residual Ground Overcurrent Elements	3
Level 1 (Amps secondary)	50G1P Level 1 (Amps secondary)	+inf A
Level 2 (Amps secondary)	50G2P Level 2 (Amps secondary)	2,50 A
Level 3 (Amps secondary)	50G3P Level 3 (Amps secondary)	2,50 A
Residual Ground Definite-Time Overcurrent Elements	Residual Ground Definite-Time Overcurrent Elements	
Level 1 (cycles in 0.25 increments)	67G1D Level 1 (cycles in 0.25 increments)	0,00 cy
Level 2 (cycles in 0.25 increments)	67G2D Level 2 (cycles in 0.25 increments)	90,00 cy
Level 3 (cycles in 0.25 increments)	67G3D Level 3 (cycles in 0.25 increments)	0,00 cy

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Negative-Sequence Instantaneous Overcurrent

Descrição	Nome	Valor
Enable Negative-Sequence Overcurrent Elements	E50Q Enable Negative-Sequence Overcurrent Elements	N
Residual Ground Definite-Time Overcurrent Elements	Negative-Sequence Definite-Time Overcurrent Elements	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Phase Time-Overcurrent

Descrição	Nome	Valor
Enable Phase Time-Overcurrent Elements	E51P Enable Phase Time-Overcurrent Elements	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Residual Ground Time-Overcurrent

Descrição	Nome	Valor
Enable Residual Ground Time-Overcurrent Elements	E51G Enable Residual Ground Time-Overcurrent Elements	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Negative-Sequence Time-Overcurrent

Descrição	Nome	Valor
Enable Negative-Sequence Time-Overcurrent Elements	E51Q Enable Negative-Sequence Time-Overcurrent Elements	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Out of Step

Descrição	Nome	Valor
Enable Out of Step Elements	EOOS Enable Out of Step Elements	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Load-Encroachment

Descrição	Nome	Valor
Enable Load-Encroachment Element	ELOAD Enable Load-Encroachment Element	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Directional Elements

Descrição	Nome	Valor
E32 Enable Directional Elements	E32 Enable Directional Elements	Y
Directional Elements	Directional Elements	
ELOP Loss of Potential Enable	ELOP Loss of Potential Enable	Y
Busbar PT LOP Logic Enable	EBBPT Busbar PT LOP Logic Enable	N
Level3 Direction	DIR3 Level3 Direction	R
Level4 Direction	DIR4 Level4 Direction	F
Ground Directional Element Priority	ORDER Ground Directional Element Priority	Q
Forward Dir Z2 Threshold (Ohms secondary)	Z2F Forward Dir Z2 Threshold (Ohms secondary)	-0,10 Ohm
Reverse Dir Z2 Threshold (Ohms secondary)	Z2R Reverse Dir Z2 Threshold (Ohms secondary)	0,10 Ohm
Forward Dir. 3I2 Pickup (Amps secondary)	50QFP Forward Dir. 3I2 Pickup (Amps secondary)	0,50 A
Reverse Dir. 3I2 Pickup (Amps secondary)	50QRP Reverse Dir. 3I2 Pickup (Amps secondary)	0,25 A
Pos.-Seq. Restraint Factor, I2/I1 (unitless)	a2 Pos.-Seq. Restraint Factor, I2/I1 (unitless)	0,10
Zero-Seq. Restraint Factor, I2/I0 (unitless)	k2 Zero-Seq. Restraint Factor, I2/I0 (unitless)	0,20
Forward Dir 3I0 Pickup (Amps secondary)	50GFP Forward Dir 3I0 Pickup (Amps secondary)	0,50 A
Reverse Dir 3I0 Pickup (Amps secondary)	50GRP Reverse Dir 3I0 Pickup (Amps secondary)	0,25 A
Pos.-Seq Restraint Factor, I0/I1 (unitless)	a0 Pos.-Seq Restraint Factor, I0/I1 (unitless)	0,10
Forward Dir. Z0 Threshold (Ohms secondary)	Z0F Forward Dir. Z0 Threshold (Ohms secondary)	1,15 Ohm
Reverse Dir. Z0 Threshold (Ohms secondary)	Z0R Reverse Dir. Z0 Threshold (Ohms secondary)	1,25 Ohm

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Voltage Elements

Descrição	Nome	Valor
Enable Voltage Elements	EVOLT Enable Voltage Elements	Y
Phase Undervoltage Elements	Phase Undervoltage Elements	
Phase Undervoltage Pickup (Volts secondary)	27P Phase Undervoltage Pickup (Volts secondary)	33,20 V
Phase Overvoltage Elements	Phase Overvoltage Elements	
Phase Overvoltage Pickup (Volts secondary)	59P Phase Overvoltage Pickup (Volts secondary)	79,67 V
Sequence Overvoltage Elements	Sequence Overvoltage Elements	
Zero-Seq. (3V0) Overvoltage Pickup (Volts secondary)	59N1P Zero-Seq. (3V0) Overvoltage Pickup (Volts secondary)	+inf V
Zero-Seq. (3V0) Overvoltage Pickup (Volts secondary)	59N2P Zero-Seq. (3V0) Overvoltage Pickup (Volts secondary)	+inf V
Neg.-Seq. (V2) Overvoltage Pickup (Volts secondary)	59QP Neg.-Seq. (V2) Overvoltage Pickup (Volts secondary)	+inf V
Pos.-Seq. (V1) Overvoltage Pickup (Volts secondary)	59V1P Pos.-Seq. (V1) Overvoltage Pickup (Volts secondary)	+inf V
Channel VS Voltage Elements	Channel VS Voltage Elements	
Channel VS Undervoltage Pickup (Volts secondary)	27SP Channel VS Undervoltage Pickup (Volts secondary)	+inf V
Channel VS Overvoltage Pickup (Volts secondary)	59SP Channel VS Overvoltage Pickup (Volts secondary)	+inf V
Phase-to-Phase Voltage Elements	Phase-to-Phase Voltage Elements	
Phase-Phase Undervoltage Pickup (Volts secondary)	27PP Phase-Phase Undervoltage Pickup (Volts secondary)	+inf V
Phase-Phase Overvoltage Pickup (Volts secondary)	59PP Phase-Phase Overvoltage Pickup (Volts secondary)	+inf V

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Synchronism Check

Descrição	Nome	Valor
Enable Synchronism Check Elements	E25 Enable Synchronism Check Elements	N
Synchronism Check Elements	Synchronism Check Elements	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Frequency Elements

Descrição	Nome	Valor
Enable Frequency Elements	E81 Enable Frequency Elements	N
Level 1	Level 1	
Level 2	Level 2	
Level 3	Level 3	
Level 4	Level 4	
Level 5	Level 5	
Level 6	Level 6	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Reclosing Relay

Descrição	Nome	Valor
Enable Reclosing Relay Elements	E79 Enable Reclosing Relay Elements	N
Reclosing Relay Elements	Reclosing Relay Elements	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Switch-Onto-Fault

Descrição	Nome	Valor
Enable Switch-Onto-Fault Elements	ESOTF Enable Switch-Onto-Fault Elements	Y
Switch-Onto-Fault Elements	Switch-Onto-Fault Elements	
Close Enable Time Delay (cycles in 0.25 increments)	CLOEND Close Enable Time Delay (cycles in 0.25 increments)	60,00 cy
52A Enable Time Delay (cycles in 0.25 increments)	52AEND 52A Enable Time Delay (cycles in 0.25 increments)	+inf cy
SOTF Duration (cycles in 0.25 increments)	SOTFD SOTF Duration (cycles in 0.25 increments)	30,00 cy

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Comm. Assisted Trip Schemes

Descrição	Nome	Valor
Enable Communication Assisted Tripping Schemes	ECOMM Enable Communication Assisted Tripping Schemes	POTT
POTT/DCUB Trip Scheme Settings	POTT/DCUB Trip Scheme Settings	
Zone 3 Reverse Block Time Delay (cycles in 0.25 increments)	Z3RBD Zone 3 Reverse Block Time Delay (cycles in 0.25 increments)	5,00 cy
Echo Block Time Delay (cycles in 0.25 increments)	EBLKD Echo Block Time Delay (cycles in 0.25 increments)	10,00 cy
Echo Time Delay Pickup (cycles in 0.25 increments)	ETDPU Echo Time Delay Pickup (cycles in 0.25 increments)	2,00 cy
Echo Duration Time Delay (cycles in 0.25 increments)	EDURD Echo Duration Time Delay (cycles in 0.25 increments)	4,00 cy
Weak-infeed enable	EWFC Weak-infeed enable	Y
WIF Phase-Phase Undervoltage (Volts secondary)	27PPW WIF Phase-Phase Undervoltage (Volts secondary)	69,00 V
WIF Zero Sequence Overvoltage (Volts secondary)	59NW WIF Zero Sequence Overvoltage (Volts secondary)	6,60 V
DCB Trip Scheme Settings	DCB Trip Scheme Settings	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Zone1 Extension

Descrição	Nome	Valor
EZ1EXT Enable Zone1 Extension Elements	EZ1EXT Enable Zone1 Extension Elements	N
Zone 1 Extension Elements	Zone 1 Extension Elements	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Demand Metering

Descrição	Nome	Valor
Enable Demand Metering Method	EDEM Enable Demand Metering Method	THM
Demand Metering Elements	Demand Metering Elements	
Time Constant	DMTC Time Constant	60
Phase Pickup (Amps secondary)	PDEMP Phase Pickup (Amps secondary)	+inf A
Residual Ground Pickup (Amps secondary)	GDEMP Residual Ground Pickup (Amps secondary)	+inf A
Negative-Sequence Pickup (Amps secondary)	QDEMP Negative-Sequence Pickup (Amps secondary)	+inf A

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Other Settings

Descrição	Nome	Valor
Minimum Trip Duration Time (cycles in 0.25 increments)	TDURD Minimum Trip Duration Time (cycles in 0.25 increments)	6,00 cy
Trip Open Pole Dropout Delay (cycles in 0.25 increments)	TOPD Trip Open Pole Dropout Delay (cycles in 0.25 increments)	2,00 cy
Close Failure Time Delay (cycles in 0.25 increments)	CFD Close Failure Time Delay (cycles in 0.25 increments)	60,00 cy
3-Pole Open Time Delay (cycles in 0.25 increments)	3POD 3-Pole Open Time Delay (cycles in 0.25 increments)	0,50 cy
Open Pole Option	OPO Open Pole Option	52
Three Pole Open Undervoltage Pickup (Volts secondary)	27PO Three Pole Open Undervoltage Pickup (Volts secondary)	40,00 V
Load Detection Phase Pickup (Amps secondary)	50LP Load Detection Phase Pickup (Amps secondary)	0,25 A

SEL 311L \ Relay Parameter Section \ Group 1 \ Logic 1 \ Trip/Comm.-Assisted Trip Logic

Descrição	Nome	Valor
Trip Logic Equations	Trip Logic Equations TR Direct Trip conditions TRCOMM Communication-assisted trip conditions TRSOTF Switch-onto-fault trip conditions DTT Direct transfer trip conditions. ULTR Unlatch trip conditions	SV2T+M1P+Z1G+Z2 T+Z3T+Z4T+67P2T+ 67G2T+TRIP87+/SV7 T+SV5T M2P+Z2G M2P+Z2G+50P3+50 G3 SV5T !50L

SEL 311L \ Relay Parameter Section \ Global \ General

Descrição	Nome	Valor
Setting Group Change Delay	TGR Setting Group Change Delay	1800,00 cy
Nominal Frequency (Hz)	NFREQ Nominal Frequency (Hz)	60
Phase Rotation	PHROT Phase Rotation	ABC
Date Format	DATE_F Date Format	YMD
DC Battery LO Voltage Pickup (Vdc)	DCLOP DC Battery LO Voltage Pickup (Vdc)	+inf V
DC Battery HI Voltage Pickup (Vdc)	DCHIP DC Battery HI Voltage Pickup (Vdc)	+inf V

SEL 311L \ Relay Parameter Section \ Channel X

Descrição	Nome	Valor
Channel X Address Check	EADDCX Channel X Address Check	Y
Channel X Transmit Address	TA_X Channel X Transmit Address	1
Channel X Receive Address	RA_X Channel X Receive Address	2
Continuous Dropout Alarm (Seconds)	RBADXP Continuous Dropout Alarm (Seconds)	1 s
Packets Lost in Last 10,000 Alarm	AVAXP Packets Lost in Last 10,000 Alarm	10
One Way Channel Delay Alarm (msec)	DBADXP One Way Channel Delay Alarm (msec)	10
Timing Source (I=Internal; E= External)	TIMRX Timing Source (I=Internal; E= External)	E

SEL 311L \ Relay Parameter Section \ Channel Y

Descrição	Nome	Valor
Channel Y Address Check	EADDCY Channel Y Address Check	Y
Channel Y Transmit Address	TA_Y Channel Y Transmit Address	3
Channel Y Receive Address	RA_Y Channel Y Receive Address	4
Continuous Dropout Alarm (Seconds)	RBADYP Continuous Dropout Alarm (Seconds)	1 s
Packets Lost in Last 10,000 Alarm	AVAYP Packets Lost in Last 10,000 Alarm	10
One Way Channel Delay Alarm (msec)	DBADYP One Way Channel Delay Alarm (msec)	10
Timing Source (I=Internal; E= External)	TIMRY Timing Source (I=Internal; E= External)	I

SEL 311L \ Additional Information \ General - General Information

Descrição	Nome	Valor
Manufacturer	Manufacturer	SEL
Device Model	Device Model	SEL 311L
Serial Number	Serial Number	serial no.
Device Address	Device Address	device address
Device type	Device type	Line Current Differential Protection
Manufacturer HW code	Hardware revision	HW revision
Software revision	Software revision	Software revision
Substation	Substation	Asa Branca
Substation Address	Substation Address	Substation address
Feeder	Feeder	feeder
Feeder Address	Feeder Address	feeder address
Bay	Bay	LT SAB
Bay address	Bay address	bay address
Protected object name	Protected object name	UPPL1
Description	Description	information
Location	Location	Location

SEL 311L \ Additional Information \ Power System Data \ VT's

Descrição	Nome	Valor
Secondary Nominal Voltage L-N	Secondary Nominal Voltage L-N	63,50 V

SEL 311L \ Additional Information \ Power System Data \ CT's

Descrição	Nome	Valor
Secondary Nominal Current	Secondary Nominal Current	5A

SEL 311L \ Additional Information \ Configuration details \ Trip logic

Descrição	Nome	Valor
Set Trip Logic Manually	Manual Trip Logic	false

SEL 311L \ Additional Information \ Configuration details \ Trip logic \ Differential

Descrição	Nome	Valor
87 Current differential elements	87 Current differential elements	true

SEL 311L \ Additional Information \ Configuration details \ Trip logic \ Distance protection

Descrição	Nome	Valor
M1P Phase distance Zone 1 used for direct tripping	M1P Phase distance Zone 1 used for direct tripping	true
Z1G Ground distance Zone 1 used for direct tripping	Z1G Ground distance Zone 1 used for direct tripping	true
M2P Phase distance Zone 2 used for direct tripping	M2P Phase distance Zone 2 used for direct tripping	true
Z2G Ground distance Zone 2 used for direct tripping	Z2G Ground distance Zone 2 used for direct tripping	true
M3P Phase distance Zone 3 used for direct tripping	M3P Phase distance Zone 3 used for direct tripping	true
Z3G Ground distance Zone 3 used for direct tripping	Z3G Ground distance Zone 3 used for direct tripping	true
M4P Phase distance Zone 4 used for direct tripping	M4P Phase distance Zone 4 used for direct tripping	true
Z4G Ground distance Zone 4 used for direct tripping	Z4G Ground distance Zone 4 used for direct tripping	true
Instantaneous trip of phase distance Zone 1	Instantaneous trip of phase distance Zone 1	true
Instantaneous trip of ground distance Zone 1	Instantaneous trip of ground distance Zone 1	true
Timer selection for phase and ground distance Zone 2	Timer selection for phase and ground distance Zone 2	Common timer for phase and ground
Timer selection for phase and ground distance Zone 3	Timer selection for phase and ground distance Zone 3	Common timer for phase and ground
Timer selection for phase and ground distance Zone 4	Timer selection for phase and ground distance Zone 4	Common timer for phase and ground

SEL 311L \ Additional Information \ Configuration details \ Trip logic \ Overcurrent

Descrição	Nome	Valor
50P1 Phase overcurrent level 1	50P1 Phase overcurrent level 1	Not used for tripping
50P2 Phase overcurrent level 2	50P2 Phase overcurrent level 2	Definite time
50P3 Phase overcurrent level 3	50P3 Phase overcurrent level 3	Not used for tripping
50G1 Residual overcurrent level 1	50G1 Residual overcurrent level 1	Not used for tripping
50G2 Residual overcurrent level 2	50G2 Residual overcurrent level 2	Definite time Directional
50G3 Residual overcurrent level 3	50G3 Residual overcurrent level 3	Not used for tripping
50G4 Residual overcurrent level 4	50G4 Residual overcurrent level 4	Not used for tripping
50Q1 Negative sequence overcurrent level 1	50Q1 Negative sequence overcurrent level 1	Not used for tripping
50Q2 Negative sequence overcurrent level 2	50Q2 Negative sequence overcurrent level 2	Not used for tripping
50Q3 Negative sequence overcurrent level 3	50Q3 Negative sequence overcurrent level 3	Not used for tripping
50Q4 Negative sequence overcurrent level 4	50Q4 Negative sequence overcurrent level 4	Not used for tripping
51PT Phase time overcurrent	51PT Phase time overcurrent	false
51GT Residual time overcurrent	51GT Residual time overcurrent	false
51QT Negative sequence time overcurrent	51QT Negative sequence time overcurrent	false

SEL 311L \ Additional Information \ Line Current Differential

Descrição	Nome	Valor
Template is Master Template	Template is Master Template	true

SEL 311L \ Additional Information \ Relay Information \ Output contacts

Descrição	Nome	Valor
Contact operatating time	Contact operatating time	4,00 ms

SEL 311L \ Additional Information \ Relay Information \ Distance Protection - Distance protection

Descrição	Nome	Valor
Rel. Tol Z at Line Angle (SIR<=60)	Rel. Tol Z	5,00 %
Max. Trip time (according operating time curve at SIR=30)	Max. operating time (cycle) - internal operation	2,00 cy
Max. Trip time (according operating time curve at SIR=30)	Max. operating time (ms) - internal operation	33,33 ms
Rel. time tolerance	Rel. t-Tolerance	5,00 %
Pick up / drop off ratio	Pick Up / Drop Off Ratio	0,95
Abs. t-Tolerance	Abs. t-Tolerance- internal operation	1,50 cy

SEL 311L \ Additional Information \ Relay Information \ Distance Protection \ Fault Detectors

Descrição	Nome	Valor
	Rel. I-Tolerance	3,00 %
	Abs. I-Tolerance	0,05 A

SEL 311L \ Additional Information \ Relay Information \ Overcurrent Protection \ Time Overcurrent Elements

Descrição	Nome	Valor
	abs. current tolerance	0,05 A
	rel. current tolerance	3,00 %
	abs. time tolerance	1,75 cy
	rel. time tolerance	0,10 %
	Pickup/Drop off ratio	1,00

SEL 311L \ Additional Information \ Relay Information \ Overcurrent Protection \ Definite Time/Instantaneous Elements

Descrição	Nome	Valor
	abs. current tolerance	0,05 A
	rel. current tolerance	3,00 %
	abs. time tolerance	1,75 cy
	rel. time tolerance	0,10 %
	Pickup/Drop off ratio	1,00
	Max operating time - internal operation	1,50 cy

SEL 311L \ Additional Information \ Relay Information \ Differential Protection

Descrição	Nome	Valor
	87L relative Tolerance	3,00 %
	87L absolute Tolerance	5,00 mA
	Angle Tolerance	3 °
	87L Trip Time	40,00 ms
	87L absolute Time Tolerance	25,00 ms

SEL 311L \ Additional Information \ Circuit Breaker Information

Descrição	Nome	Valor
52ab Percentage	52ab Percentage	20,00 %
CB Trip Time	CB Trip Time	50,00 ms
CB Close Time	CB Close Time	100,00 ms
t-Tolerance abs	t-Tolerance abs	0,00 ms
t-Tolerance rel	t-Tolerance rel	5,00 %

SEL 311L \ Additional Information \ Test Tolerance

Descrição	Nome	Valor
Global Test Tolerance	Test Tolerance	2,00 %

Objeto sob Teste - Outras Funções RIO

CB Configuration

Descrição	Nome	Valor
Tempo de disparo do DJ	Tempo de disparo do DJ	50,00 ms
Tempo de fechamento do DJ	Tempo de fechamento do DJ	100,00 ms
Tempos para 52a, 52b em porcentagem de tempo CB	52a, 52b % de DJ	20,00 %

Equipamento em teste - Parâmetros de Sobrecorrente

Valores - Gerais:

TolAbsdeTempo:	0,03 s	Conexão de TP:	No objeto protegido
TolReldeTempo:	0,10 %	Conexão de ponto neutro do TP:	Para objeto protegido
TolAbsdeCorrente:	0,05 Iref		
TolReldeCorrente:	3,00 %		
Direcional:	Sim		

Elementos - Fase:

Ativo	Nome	Característica de Disparo	I Pick-up	Tempo	Relação de Reset	Direção
Não	51P	U3 Very Inverse	0,10 Iref	2,00	1,00	Não Direcional
Não	50/67P1	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Sim	50/67P2	Definite Time	5,00 Iref	1,53 s	1,00	Não Direcional
Não	50/67P3	Definite Time	28,00 Iref	0,03 s	1,00	Não Direcional

Elementos - Residual:

Ativo	Nome	Característica de Disparo	I Pick-up	Tempo	Relação de Reset	Direção
Não	51G	U3 Very Inverse	0,10 Iref	2,00	1,00	Para frente
Não	50/67G1	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Sim	50/67G2	Definite Time	2,50 Iref	1,53 s	1,00	Para frente
Não	50/67G3	Definite Time	2,50 Iref	0,03 s	1,00	Não Direcional
Não	50/67G4	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional

Elementos - Sequência Negativa:

Ativo	Nome	Característica de Disparo	I Pick-up	Tempo	Relação de Reset	Direção
Não	51Q	U3 Very Inverse	0,10 Iref	2,00	1,00	Não Direcional
Não	50/67Q1	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Não	50/67Q2	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Não	50/67Q3	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Não	50/67Q4	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional

Equipamento em teste - Ajustes de Distância

Parâmetros do sistema:

Comprimento da linha:	1,230 Ω	Ângulo da linha:	79,39 °
Conexão TP:	na linha	Ponto de Neutro do TC:	Dir.linha
Correção da impedância 1A/I nom:	Não		
Impedância em valores primários:	Não		

Tolerâncias:

Tol. T rel.:	5,000 %	Tol. T abs. -:	29,00 ms
Tol. T abs. +:	29,00 ms	Tol. Z abs.:	50,00 mΩ
Tol. rel. Z:	5,000 %		

Fator de terra:

Mag. kL:

0,000000

Ângulo kL:

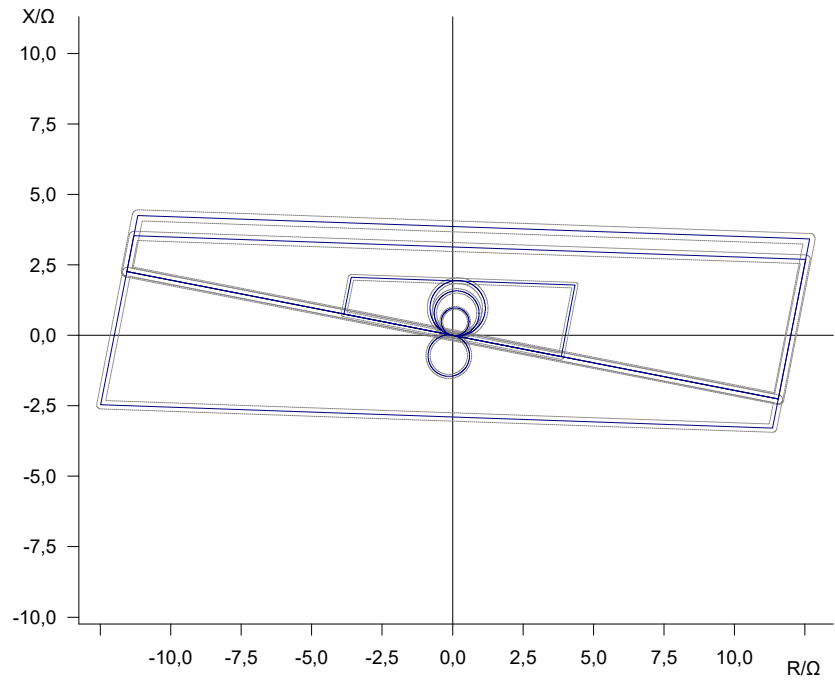
0,000000°

Separar a resistência de arco:

Não

Ajustes de zona:

Etiqueta	Tipo	Loop de Falta	Tempo de Disparo	Tol.T rel	Tol.T abs+	Tol.T abs-	Tol.Z rel.	Tol.Z abs
ZL1	Disparo	L-L	33,33 ms	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 mΩ
ZL2	Disparo	L-L	433,3 ms	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 mΩ
ZL3	Disparo	L-L	1,533 s	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 mΩ
ZL4	Disparo	L-L	1,533 s	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 mΩ
ZX1	Disparo	L-N	33,33 ms	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 mΩ
ZX2	Disparo	L-N	433,3 ms	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 mΩ
ZX3	Disparo	L-N	1,533 s	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 mΩ
ZX4	Disparo	L-N	1,533 s	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 mΩ



Objeto sob teste - Configurações de potência

Parâmetros do sistema:

Ponto de Neutro do TC:

Em direção à linha

Referência para dados de potência relativa:

Medidor:

TC*VT nominal

Potência nominal de referência:

0,00 VA

Tolerâncias:

Tol. T rel.:

5,000 %

Tol. S rel.:

5,000 %

Tol. T abs. +:

50,00 ms

Tol. S abs.:

50,00 mVA

Tol. T abs. -:

0,00 s

Tol. de ângulo:

3,000 °

Limiar de tensão:

Limiar de tensão ativado:

Não

Tol. rel. limiar de tensão:

5,000 %

Limiar de tensão:

100,0 V

Tol. abs. limiar de tensão:

5,000 V

Tipo de limiar de tensão:

Subtensão

Limiar de sobrecorrente:

Limiar de corrente
ativado: Não
Limiar de
sobrecorrente: 1,000 A

Tol. rel. limiar de
sobrecorrente: 5,000 %
Tol. abs. limiar de
sobrecorrente: 250,0 mA

Ajustes de zona:**Hardware Configuration****Equipamento de Teste**

Tipo	No de série
CMC 356	

Verificação de Hardware

Executado Por	Resultado	Detalhes
Ainda não executado		

Saídas analógicas

Equipamento de Teste		Equipamento em teste		
Dispositivo	Conector	Etiqueta	Terminal de conexão	
CMC 356 V	1	V A-N		
	2	V B-N		
	3	V C-N		
	N			
CMC 356 V (4)	1	V(2)-1		
	N			
CMC 356 I (A)	1	I A		
	2	I B		
	3	I C		
	N			
CMC 356 I (B)	1	I(2)-1		
	2	I(2)-2		
	3	I(2)-3		
	N			

Entradas binárias / analógicas

Equipamento de Teste		Equipamento em teste		
Dispositivo	Conector	Etiqueta	Terminal de conexão	
CMC356	1+	Entr. bin. 1		
	1-			
	2+	Entr. bin. 2		
	2-			
	3+	Entr. bin. 3		
	3-			
	4+	Entr. bin. 4		
	4-			
	5+	Entr. bin. 5		
	5-			
	6+	Entr. bin. 6		
	6-			
	7+	Entr. bin. 7		
	7-			
	8+	Entr. bin. 8		
	8-			
	9+	Entr. bin. 9		
	9-			
	10+	Entr. bin. 10		
	10-			
	1	Entr. bin. 11		
	2	Entr. bin. 12		
	N			

Saídas binárias

Equipamento de Teste		Equipamento em teste		
Dispositivo	Conector	Etiqueta	Terminal de conexão	
CMC356	1+	Saída bin. 1		
	1-			
	2+	Saída bin. 2		
	2-			
	3+	Saída bin. 3		
	3-			
	4+	Saída bin. 4		
	4-			
	11	Saída bin. 5		
	12	Saída bin. 6		
	13	Saída bin. 7		
	14	Saída bin. 8		
	N			

Entradas analógicas DC

Equipamento de Teste		Equipamento em teste		
Dispositivo	Conector	Etiqueta	Terminal de conexão	
CMC 356	V+	V1		
	V-			
	I+	I1		
	I-			

Sampled Values

Visão geral

Stream	Habilitar	IED / Sampled Values ID	Simulado	Porta Ethernet
1	Não		Sim	ETH2
2	Não		Sim	ETH2
3	Não		Sim	ETH2

Taxa de amostragem/número de ASDUs:

n/a

Advanced Distance:

Equipamento em Teste - Ajustes do dispositivo

Subestação/Vão:

Subestação: Asa Branca
Bay: LT SAB

Endereço da Subestação: Substation address
Endereço do Vão: bay address

Dispositivo:

Nome/descrição: SEL 311L
Tipo de Dispositivo: Line Current Differential Protection
Número Modelo/Série: serial no.
Info adicional 1: UPPL1
Info adicional 2: feeder address

Fabricante: SEL
Endereço do dispositivo: device address

Valores Nominais:

f nom: 60,00 Hz
V nom (secundária): 110,0 V
I nom (secundária): 5,000 A

Número de fases: 3
V primária: 132,0 kV
I primária: 1,200 kA

Fator para corrente/tensão residual:

VLN / VN: 1,732
VN (secundário): 36,66 V
Direção da tensão residual: 3 * V0

IN / I nom: 1,000
IN (secundário): 1,000 A
Direção da corrente residual: -3 * IO

Limites:

V max: 500,0 V

I max: 250,0 A

Filtros Debounce/Deglitch:

T de Debounce: 3,000 ms

T de Deglitch: 0,00 s

Detecção de Sobrecarga:

Tempo de Supressão: 50,00 ms

Outras propriedades do equipamento:

Tempo de queda: 20,00 ms

Objeto sob teste - Ajustes do usuário

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ General Settings

Descrição	Nome	Valor
Relay Identifier Lables	Relay Identifier Lables	
Relay Identifier (30 chars)	RID Relay Identifier (30 chars)	UPPL1_LT JOAO CAMARA III
Terminal Identifier (30 chars)	TID Terminal Identifier (30 chars)	UEE ASA BRANCA
Current Transformer Ratio and Application Settings	Current Transformer Ratio and Application Settings	
Local Phase (IA,IB,IC) CT Ratio, CTR:1	CTR Local Phase (IA,IB,IC) CT Ratio, CTR:1	240
Application	APP Application	311L
Advance Setting Enable	EADVS Advance Setting Enable	Y

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Line Current Differential Settings

Descrição	Nome	Valor
Line Current Differential Configuration Settings	Line Current Differential Configuration Settings	
Number of 87L Terminals	E87L Number of 87L Terminals	2
Enable High Speed Tripping	EHST Enable High Speed Tripping	N
Enable High Speed Direct Transfer Trip	EHSDDT Enable High Speed Direct Transfer Trip	Y
Enable Delta Current Detect	EDD Enable Delta Current Detect	Y
ETAP Tapped Load Coordination	ETAP Tapped Load Coordination	N
Enable Open CT Logic	EOCTL Enable Open CT Logic	N
Primary 87L Channel	PCHAN Primary 87L Channel	X
Hot-Standby Channel Feature	EHSC Hot-Standby Channel Feature	N
CTR at Terminal Connected to Channel X	CTR_X CTR at Terminal Connected to Channel X	240

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Line Current Differential Settings \ 87L Settings

Descrição	Nome	Valor
Minimum Difference Current Enable Level Settings	Minimum Difference Current Enable Level Settings	
Phase 87L (Amps secondary)	87LPP Phase 87L (Amps secondary)	6,00 A
Negative-Sequence 87L (Amps secondary)	87L2P Negative-Sequence 87L (Amps secondary)	0,50 A
Ground 87L (Amps secondary)	87LGP Ground 87L (Amps secondary)	+inf A
Ph. Diff. Current Alarm Pickup (Amps secondary)	CTALARM Ph. Diff. Current Alarm Pickup (Amps secondary)	0,50 A
Restraint Region Characteristic Settings	Restraint Region Characteristic Settings	
Outer Radius	87LR Outer Radius	6,0
Angle (degrees)	87LANG Angle (degrees)	195 °

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Backup Protection and Line Parameters

Descrição	Nome	Valor
Backup Protection Transformer Ratio Settings	Backup Protection Transformer Ratio Settings	
Polarizing (IPOL) CT Ratio, CTRP:1	CTRP Polarizing (IPOL) CT Ratio, CTRP:1	200
Phase (VA,VB,VC) PT Ratio, PTR:1	PTR Phase (VA,VB,VC) PT Ratio, PTR:1	1200,00
Synch. Voltage (VS) PT Ratio, PTRS:1	PTRS Synch. Voltage (VS) PT Ratio, PTRS:1	1200,00
Line Settings	Line Settings	
Pos-Seq Line Impedance Magnitude (Ohms secondary)	Z1MAG Pos-Seq Line Impedance Magnitude (Ohms secondary)	1,23 Ohm
Pos-Seq Line Impedance Angle (degrees)	Z1ANG Pos-Seq Line Impedance Angle (degrees)	79,39 °
Zero-Seq Line Impedance Magnitude (Ohms secondary)	Z0MAG Zero-Seq Line Impedance Magnitude (Ohms secondary)	4,89 Ohm
Zero-Seq Line Impedance Angle (degrees)	Z0ANG Zero-Seq Line Impedance Angle (degrees)	78,68 °
Line Length (unitless)	LL Line Length (unitless)	13,10
Fault Locator	Fault Locator	
Fault Locator Enable	EFLOC Fault Locator Enable	Y

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Phase Distance

Descrição	Nome	Valor
Enable Mho Phase Distance Elements	E21P Enable Mho Phase Distance Elements	4
CCVT Transient Detection Enable	ECCVT CCVT Transient Detection Enable	N
Mho Phase Distance Element Reach Settings	Mho Phase Distance Element Reach Settings	
Reach Zone1 (Ohm secondary)	Z1P Reach Zone1 (Ohm secondary)	0,98 Ohm
Reach Zone2 (Ohm secondary)	Z2P Reach Zone2 (Ohm secondary)	1,59 Ohm
Reach Zone3 (Ohm secondary)	Z3P Reach Zone3 (Ohm secondary)	1,47 Ohm
Reach Zone4 (Ohm secondary)	Z4P Reach Zone4 (Ohm secondary)	1,96 Ohm
Mho Phase Distance Overcurrent Fault Detector Settings	Mho Phase Distance Overcurrent Fault Detector Settings	
Phase-Phase Overcurrent Fault Detector Zone1 (Amps secondary)	50PP1 Phase-Phase Overcurrent Fault Detector Zone1 (Amps secondary)	0,50 A
Phase-Phase Overcurrent Fault Detector Zone2 (Amps secondary)	50PP2 Phase-Phase Overcurrent Fault Detector Zone2 (Amps secondary)	0,50 A
Phase-Phase Overcurrent Fault Detector Zone3 (Amps secondary)	50PP3 Phase-Phase Overcurrent Fault Detector Zone3 (Amps secondary)	0,50 A
Phase-Phase Overcurrent Fault Detector Zone4 (Amps secondary)	50PP4 Phase-Phase Overcurrent Fault Detector Zone4 (Amps secondary)	0,50 A

Descrição	Nome	Valor
Enable Mho Ground Distance Elements	E21MG Enable Mho Ground Distance Elements	N
E21XG Enable Quad Ground Distance Elements	E21XG Enable Quad Ground Distance Elements	
Enable Quad Ground Distance Elements	E21XG Enable Quad Ground Distance Elements	4
Reactance Zone1 Quad Ground (Ohm secondary)	XG1 Quad Zone1 Reactance (Ohm secondary)	0,98 Ohm
Reactance Zone2 Quad Ground (Ohm secondary)	XG2 Quad Zone2 Reactance (Ohm secondary)	1,59 Ohm
Reactance Zone3 Quad Ground (Ohm secondary)	XG3 Quad Zone3 Reactance (Ohm secondary)	1,47 Ohm
Reactance Zone4 Quad Ground (Ohm secondary)	XG4 Quad Zone4 Reactance (Ohm secondary)	1,96 Ohm
Resistance Zone1 Quad Ground (Ohm secondary)	RG1 Quad Zone1 Resistance (Ohm secondary)	4,00 Ohm
Resistance Zone2 Quad Ground (Ohm secondary)	RG2 Quad Zone2 Resistance (Ohm secondary)	12,00 Ohm
Resistance Zone3 Quad Ground (Ohm secondary)	RG3 Quad Zone3 Resistance (Ohm secondary)	12,00 Ohm
Resistance Zone4 Quad Ground (Ohm secondary)	RG4 Quad Zone4 Resistance (Ohm secondary)	12,00 Ohm
Quad Ground Polarizing Quantity	XGPOL Quad Ground Polarizing Quantity	I2
None Homogeneous Correction Angle (degrees)	TANG None Homogeneous Correction Angle (degrees)	-2,00 °
Quad and Mho Ground Distance Fault Detector Settings	Quad and Mho Ground Distance Fault Detector Settings	
Phase Overcurrent Fault Detector Zone1 for Ground Fault (Amps secondary)	50L1 Zone1 Phase Current FD (Amps secondary)	0,50 A
Phase Overcurrent Fault Detector Zone2 for Ground Fault (Amps secondary)	50L2 Zone2 Phase Current FD (Amps secondary)	0,50 A
Phase Overcurrent Fault Detector Zone3 for Ground Fault (Amps secondary)	50L3 Zone3 Phase Current FD (Amps secondary)	0,50 A
Phase Overcurrent Fault Detector Zone4 for Ground Fault (Amps secondary)	50L4 Zone4 Phase Current FD (Amps secondary)	0,50 A
Residual Current Fault Detector Zone1 for Ground Fault (Amps secondary)	50GZ1 Zone1 Residual Current FD (Amps secondary)	0,50 A
Residual Current Fault Detector Zone2 for Ground Fault (Amps secondary)	50GZ2 Zone2 Residual Current FD (Amps secondary)	0,50 A
Residual Current Fault Detector Zone3 for Ground Fault (Amps secondary)	50GZ3 Zone3 Residual Current FD (Amps secondary)	0,50 A
Residual Current Fault Detector Zone4 for Ground Fault (Amps secondary)	50GZ4 Zone4 Residual Current FD (Amps secondary)	0,50 A
Zero-Sequence Compensation Settings	Zero-Sequence Compensation Settings	
Zone1 ZSC Factor Magnitude	K0M1 Zone1 ZSC Factor Magnitude	0,995
Zone1 ZSC Factor Angle	K0A1 Zone1 ZSC Factor Angle	-0,95 °
Zone2,3,4 ZSC Factor Magnitude	K0M Zone2,3,4 ZSC Factor Magnitude	0,995
Zone2,3,4 ZSC Factor Angle	K0A Zone2,3,4 ZSC Factor Angle	-0,95 °
Mho Phase Distance Element Time Delay Settings	Mho Phase Distance Element Time Delay Settings	
Zone1 Phase Time Delay (cycles in 0.25 increments)	Z1PD Zone1 Phase Time Delay (cycles in 0.25 increments)	0,00 cy
Zone2 Phase Time Delay (cycles in 0.25 increments)	Z2PD Zone2 Phase Time Delay (cycles in 0.25 increments)	24,00 cy
Zone3 Phase Time Delay (cycles in 0.25 increments)	Z3PD Zone3 Phase Time Delay (cycles in 0.25 increments)	90,00 cy
Zone4 Phase Time Delay (cycles in 0.25 increments)	Z4PD Zone4 Phase Time Delay (cycles in 0.25 increments)	90,00 cy
Mho Ground Distance Element Time Delay Settings	Mho Ground Distance Element Time Delay Settings	

Zone1 Ground Time Delay (cycles in 0.25 increments)	Z1GD Zone1 Ground Time Delay (cycles in 0.25 increments)	0,00 cy
Zone2 Ground Time Delay (cycles in 0.25 increments)	Z2GD Zone2 Ground Time Delay (cycles in 0.25 increments)	24,00 cy
Zone3 Ground Time Delay (cycles in 0.25 increments)	Z3GD Zone3 Ground Time Delay (cycles in 0.25 increments)	90,00 cy
Zone4 Ground Time Delay (cycles in 0.25 increments)	Z4GD Zone4 Ground Time Delay (cycles in 0.25 increments)	90,00 cy
Common Phase/Ground Dist. Element Time Delay Settings	Common Phase/Ground Dist. Element Time Delay Settings	
Zone1 Common Ph/Gr Time Delay (cycles in 0.25 increments)	Z1D Zone1 Common Ph/Gr Time Delay (cycles in 0.25 increments)	0,00 cy
Zone2 Common Ph/Gr Time Delay (cycles in 0.25 increments)	Z2D Zone2 Common Ph/Gr Time Delay (cycles in 0.25 increments)	24,00 cy
Zone3 Common Ph/Gr Time Delay (cycles in 0.25 increments)	Z3D Zone3 Common Ph/Gr Time Delay (cycles in 0.25 increments)	90,00 cy
Zone4 Common Ph/Gr Time Delay (cycles in 0.25 increments)	Z4D Zone4 Common Ph/Gr Time Delay (cycles in 0.25 increments)	90,00 cy

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Phase Instantaneous Overcurrent

Descrição	Nome	Valor
Enable Phase Overcurrent Elements	E50P Enable Phase Overcurrent Elements	3
Level 1 (Amps secondary)	50P1P Level 1 (Amps secondary)	+inf A
Level 2 (Amps secondary)	50P2P Level 2 (Amps secondary)	5,00 A
Level 3 (Amps secondary)	50P3P Level 3 (Amps secondary)	28,00 A
Phase Definite-Time Overcurrent Elements	Phase Definite-Time Overcurrent Elements	
Level 1 (cycles in 0.25 increments)	67P1D Level 1 (cycles in 0.25 increments)	0,00 cy
Level 2 (cycles in 0.25 increments)	67P2D Level 2 (cycles in 0.25 increments)	90,00 cy
Level 3 (cycles in 0.25 increments)	67P3D Level 3 (cycles in 0.25 increments)	0,00 cy

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Residual Ground Instantaneous Overcurrent

Descrição	Nome	Valor
Enable Residual Ground Overcurrent Elements	E50G Enable Residual Ground Overcurrent Elements	3
Level 1 (Amps secondary)	50G1P Level 1 (Amps secondary)	+inf A
Level 2 (Amps secondary)	50G2P Level 2 (Amps secondary)	2,50 A
Level 3 (Amps secondary)	50G3P Level 3 (Amps secondary)	2,50 A
Residual Ground Definite-Time Overcurrent Elements	Residual Ground Definite-Time Overcurrent Elements	
Level 1 (cycles in 0.25 increments)	67G1D Level 1 (cycles in 0.25 increments)	0,00 cy
Level 2 (cycles in 0.25 increments)	67G2D Level 2 (cycles in 0.25 increments)	90,00 cy
Level 3 (cycles in 0.25 increments)	67G3D Level 3 (cycles in 0.25 increments)	0,00 cy

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Negative-Sequence Instantaneous Overcurrent

Descrição	Nome	Valor
Enable Negative-Sequence Overcurrent Elements	E50Q Enable Negative-Sequence Overcurrent Elements	N
Residual Ground Definite-Time Overcurrent Elements	Negative-Sequence Definite-Time Overcurrent Elements	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Phase Time-Overcurrent

Descrição	Nome	Valor
Enable Phase Time-Overcurrent Elements	E51P Enable Phase Time-Overcurrent Elements	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Residual Ground Time-Overcurrent

Descrição	Nome	Valor
Enable Residual Ground Time-Overcurrent Elements	E51G Enable Residual Ground Time-Overcurrent Elements	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Negative-Sequence Time-Overcurrent

Descrição	Nome	Valor
Enable Negative-Sequence Time-Overcurrent Elements	E51Q Enable Negative-Sequence Time-Overcurrent Elements	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Out of Step

Descrição	Nome	Valor
Enable Out of Step Elements	EOOS Enable Out of Step Elements	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Load-Encroachment

Descrição	Nome	Valor
Enable Load-Encroachment Element	ELOAD Enable Load-Encroachment Element	N

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Directional Elements

Descrição	Nome	Valor
E32 Enable Directional Elements	E32 Enable Directional Elements	Y
Directional Elements	Directional Elements	
ELOP Loss of Potential Enable	ELOP Loss of Potential Enable	Y
Busbar PT LOP Logic Enable	EBBPT Busbar PT LOP Logic Enable	N
Level3 Direction	DIR3 Level3 Direction	R
Level4 Direction	DIR4 Level4 Direction	F
Ground Directional Element Priority	ORDER Ground Directional Element Priority	Q
Forward Dir Z2 Threshold (Ohms secondary)	Z2F Forward Dir Z2 Threshold (Ohms secondary)	-0,10 Ohm
Reverse Dir Z2 Threshold (Ohms secondary)	Z2R Reverse Dir Z2 Threshold (Ohms secondary)	0,10 Ohm
Forward Dir. 3I2 Pickup (Amps secondary)	50QFP Forward Dir. 3I2 Pickup (Amps secondary)	0,50 A
Reverse Dir. 3I2 Pickup (Amps secondary)	50QRP Reverse Dir. 3I2 Pickup (Amps secondary)	0,25 A
Pos.-Seq. Restraint Factor, I2/I1 (unitless)	a2 Pos.-Seq. Restraint Factor, I2/I1 (unitless)	0,10
Zero-Seq. Restraint Factor, I2/I0 (unitless)	k2 Zero-Seq. Restraint Factor, I2/I0 (unitless)	0,20
Forward Dir 3I0 Pickup (Amps secondary)	50GFP Forward Dir 3I0 Pickup (Amps secondary)	0,50 A
Reverse Dir 3I0 Pickup (Amps secondary)	50GRP Reverse Dir 3I0 Pickup (Amps secondary)	0,25 A
Pos.-Seq Restraint Factor, I0/I1 (unitless)	a0 Pos.-Seq Restraint Factor, I0/I1 (unitless)	0,10
Forward Dir. Z0 Threshold (Ohms secondary)	Z0F Forward Dir. Z0 Threshold (Ohms secondary)	1,15 Ohm
Reverse Dir. Z0 Threshold (Ohms secondary)	Z0R Reverse Dir. Z0 Threshold (Ohms secondary)	1,25 Ohm

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Voltage Elements

Descrição	Nome	Valor
Enable Voltage Elements	EVOLT Enable Voltage Elements	Y
Phase Undervoltage Elements	Phase Undervoltage Elements	
Phase Undervoltage Pickup (Volts secondary)	27P Phase Undervoltage Pickup (Volts secondary)	33,20 V
Phase Overvoltage Elements	Phase Overvoltage Elements	
Phase Overvoltage Pickup (Volts secondary)	59P Phase Overvoltage Pickup (Volts secondary)	79,67 V
Sequence Overvoltage Elements	Sequence Overvoltage Elements	
Zero-Seq. (3V0) Overvoltage Pickup (Volts secondary)	59N1P Zero-Seq. (3V0) Overvoltage Pickup (Volts secondary)	+inf V
Zero-Seq. (3V0) Overvoltage Pickup (Volts secondary)	59N2P Zero-Seq. (3V0) Overvoltage Pickup (Volts secondary)	+inf V
Neg.-Seq. (V2) Overvoltage Pickup (Volts secondary)	59QP Neg.-Seq. (V2) Overvoltage Pickup (Volts secondary)	+inf V
Pos.-Seq. (V1) Overvoltage Pickup (Volts secondary)	59V1P Pos.-Seq. (V1) Overvoltage Pickup (Volts secondary)	+inf V
Channel VS Voltage Elements	Channel VS Voltage Elements	
Channel VS Undervoltage Pickup (Volts secondary)	27SP Channel VS Undervoltage Pickup (Volts secondary)	+inf V
Channel VS Overvoltage Pickup (Volts secondary)	59SP Channel VS Overvoltage Pickup (Volts secondary)	+inf V
Phase-to-Phase Voltage Elements	Phase-to-Phase Voltage Elements	
Phase-Phase Undervoltage Pickup (Volts secondary)	27PP Phase-Phase Undervoltage Pickup (Volts secondary)	+inf V
Phase-Phase Overvoltage Pickup (Volts secondary)	59PP Phase-Phase Overvoltage Pickup (Volts secondary)	+inf V

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Synchronism Check

Descrição	Nome	Valor
Enable Synchronism Check Elements	E25 Enable Synchronism Check Elements	N
Synchronism Check Elements	Synchronism Check Elements	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Frequency Elements

Descrição	Nome	Valor
Enable Frequency Elements	E81 Enable Frequency Elements	N
Level 1	Level 1	
Level 2	Level 2	
Level 3	Level 3	
Level 4	Level 4	
Level 5	Level 5	
Level 6	Level 6	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Reclosing Relay

Descrição	Nome	Valor
Enable Reclosing Relay Elements	E79 Enable Reclosing Relay Elements	N
Reclosing Relay Elements	Reclosing Relay Elements	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Switch-Onto-Fault

Descrição	Nome	Valor
Enable Switch-Onto-Fault Elements	ESOTF Enable Switch-Onto-Fault Elements	Y
Switch-Onto-Fault Elements	Switch-Onto-Fault Elements	
Close Enable Time Delay (cycles in 0.25 increments)	CLOEND Close Enable Time Delay (cycles in 0.25 increments)	60,00 cy
52A Enable Time Delay (cycles in 0.25 increments)	52AEND 52A Enable Time Delay (cycles in 0.25 increments)	+inf cy
SOTF Duration (cycles in 0.25 increments)	SOTFD SOTF Duration (cycles in 0.25 increments)	30,00 cy

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Comm. Assisted Trip Schemes

Descrição	Nome	Valor
Enable Communication Assisted Tripping Schemes	ECOMM Enable Communication Assisted Tripping Schemes	POTT
POTT/DCUB Trip Scheme Settings	POTT/DCUB Trip Scheme Settings	
Zone 3 Reverse Block Time Delay (cycles in 0.25 increments)	Z3RBD Zone 3 Reverse Block Time Delay (cycles in 0.25 increments)	5,00 cy
Echo Block Time Delay (cycles in 0.25 increments)	EBLKD Echo Block Time Delay (cycles in 0.25 increments)	10,00 cy
Echo Time Delay Pickup (cycles in 0.25 increments)	ETDPU Echo Time Delay Pickup (cycles in 0.25 increments)	2,00 cy
Echo Duration Time Delay (cycles in 0.25 increments)	EDURD Echo Duration Time Delay (cycles in 0.25 increments)	4,00 cy
Weak-infeed enable	EWFC Weak-infeed enable	Y
WIF Phase-Phase Undervoltage (Volts secondary)	27PPW WIF Phase-Phase Undervoltage (Volts secondary)	69,00 V
WIF Zero Sequence Overvoltage (Volts secondary)	59NW WIF Zero Sequence Overvoltage (Volts secondary)	6,60 V
DCB Trip Scheme Settings	DCB Trip Scheme Settings	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Zone1 Extension

Descrição	Nome	Valor
EZ1EXT Enable Zone1 Extension Elements	EZ1EXT Enable Zone1 Extension Elements	N
Zone 1 Extension Elements	Zone 1 Extension Elements	

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Demand Metering

Descrição	Nome	Valor
Enable Demand Metering Method	EDEM Enable Demand Metering Method	THM
Demand Metering Elements	Demand Metering Elements	
Time Constant	DMTC Time Constant	60
Phase Pickup (Amps secondary)	PDEMP Phase Pickup (Amps secondary)	+inf A
Residual Ground Pickup (Amps secondary)	GDEMP Residual Ground Pickup (Amps secondary)	+inf A
Negative-Sequence Pickup (Amps secondary)	QDEMP Negative-Sequence Pickup (Amps secondary)	+inf A

SEL 311L \ Relay Parameter Section \ Group 1 \ Set 1 \ Other Settings

Descrição	Nome	Valor
Minimum Trip Duration Time (cycles in 0.25 increments)	TDURD Minimum Trip Duration Time (cycles in 0.25 increments)	6,00 cy
Trip Open Pole Dropout Delay (cycles in 0.25 increments)	TOPD Trip Open Pole Dropout Delay (cycles in 0.25 increments)	2,00 cy
Close Failure Time Delay (cycles in 0.25 increments)	CFD Close Failure Time Delay (cycles in 0.25 increments)	60,00 cy
3-Pole Open Time Delay (cycles in 0.25 increments)	3POD 3-Pole Open Time Delay (cycles in 0.25 increments)	0,50 cy
Open Pole Option	OPO Open Pole Option	52
Three Pole Open Undervoltage Pickup (Volts secondary)	27PO Three Pole Open Undervoltage Pickup (Volts secondary)	40,00 V
Load Detection Phase Pickup (Amps secondary)	50LP Load Detection Phase Pickup (Amps secondary)	0,25 A

SEL 311L \ Relay Parameter Section \ Group 1 \ Logic 1 \ Trip/Comm.-Assisted Trip Logic

Descrição	Nome	Valor
Trip Logic Equations	Trip Logic Equations TR Direct Trip conditions TRCOMM Communication-assisted trip conditions TRSOTF Switch-onto-fault trip conditions DTT Direct transfer trip conditions. ULTR Unlatch trip conditions	SV2T+M1P+Z1G+Z2 T+Z3T+Z4T+67P2T+ 67G2T+TRIP87+/SV7 T+SV5T M2P+Z2G M2P+Z2G+50P3+50 G3 SV5T !50L

SEL 311L \ Relay Parameter Section \ Global \ General

Descrição	Nome	Valor
Setting Group Change Delay	TGR Setting Group Change Delay	1800,00 cy
Nominal Frequency (Hz)	NFREQ Nominal Frequency (Hz)	60
Phase Rotation	PHROT Phase Rotation	ABC
Date Format	DATE_F Date Format	YMD
DC Battery LO Voltage Pickup (Vdc)	DCLOP DC Battery LO Voltage Pickup (Vdc)	+inf V
DC Battery HI Voltage Pickup (Vdc)	DCHIP DC Battery HI Voltage Pickup (Vdc)	+inf V

SEL 311L \ Relay Parameter Section \ Channel X

Descrição	Nome	Valor
Channel X Address Check	EADDCX Channel X Address Check	Y
Channel X Transmit Address	TA_X Channel X Transmit Address	1
Channel X Receive Address	RA_X Channel X Receive Address	2
Continuous Dropout Alarm (Seconds)	RBADXP Continuous Dropout Alarm (Seconds)	1 s
Packets Lost in Last 10,000 Alarm	AVAXP Packets Lost in Last 10,000 Alarm	10
One Way Channel Delay Alarm (msec)	DBADXP One Way Channel Delay Alarm (msec)	10
Timing Source (I=Internal; E= External)	TIMRX Timing Source (I=Internal; E= External)	E

SEL 311L \ Relay Parameter Section \ Channel Y

Descrição	Nome	Valor
Channel Y Address Check	EADDCY Channel Y Address Check	Y
Channel Y Transmit Address	TA_Y Channel Y Transmit Address	3
Channel Y Receive Address	RA_Y Channel Y Receive Address	4
Continuous Dropout Alarm (Seconds)	RBADYP Continuous Dropout Alarm (Seconds)	1 s
Packets Lost in Last 10,000 Alarm	AVAYP Packets Lost in Last 10,000 Alarm	10
One Way Channel Delay Alarm (msec)	DBADYP One Way Channel Delay Alarm (msec)	10
Timing Source (I=Internal; E= External)	TIMRY Timing Source (I=Internal; E= External)	I

SEL 311L \ Additional Information \ General - General Information

Descrição	Nome	Valor
Manufacturer	Manufacturer	SEL
Device Model	Device Model	SEL 311L
Serial Number	Serial Number	serial no.
Device Address	Device Address	device address
Device type	Device type	Line Current Differential Protection
Manufacturer HW code	Hardware revision	HW revision
Software revision	Software revision	Software revision
Substation	Substation	Asa Branca
Substation Address	Substation Address	Substation address
Feeder	Feeder	feeder
Feeder Address	Feeder Address	feeder address
Bay	Bay	LT SAB
Bay address	Bay address	bay address
Protected object name	Protected object name	UPPL1
Description	Description	information
Location	Location	Location

SEL 311L \ Additional Information \ Power System Data \ VT's

Descrição	Nome	Valor
Secondary Nominal Voltage L-N	Secondary Nominal Voltage L-N	63,50 V

SEL 311L \ Additional Information \ Power System Data \ CT's

Descrição	Nome	Valor
Secondary Nominal Current	Secondary Nominal Current	5A

SEL 311L \ Additional Information \ Configuration details \ Trip logic

Descrição	Nome	Valor
Set Trip Logic Manually	Manual Trip Logic	false

SEL 311L \ Additional Information \ Configuration details \ Trip logic \ Differential

Descrição	Nome	Valor
87 Current differential elements	87 Current differential elements	true

SEL 311L \ Additional Information \ Configuration details \ Trip logic \ Distance protection

Descrição	Nome	Valor
M1P Phase distance Zone 1 used for direct tripping	M1P Phase distance Zone 1 used for direct tripping	true
Z1G Ground distance Zone 1 used for direct tripping	Z1G Ground distance Zone 1 used for direct tripping	true
M2P Phase distance Zone 2 used for direct tripping	M2P Phase distance Zone 2 used for direct tripping	true
Z2G Ground distance Zone 2 used for direct tripping	Z2G Ground distance Zone 2 used for direct tripping	true
M3P Phase distance Zone 3 used for direct tripping	M3P Phase distance Zone 3 used for direct tripping	true
Z3G Ground distance Zone 3 used for direct tripping	Z3G Ground distance Zone 3 used for direct tripping	true
M4P Phase distance Zone 4 used for direct tripping	M4P Phase distance Zone 4 used for direct tripping	true
Z4G Ground distance Zone 4 used for direct tripping	Z4G Ground distance Zone 4 used for direct tripping	true
Instantaneous trip of phase distance Zone 1	Instantaneous trip of phase distance Zone 1	true
Instantaneous trip of ground distance Zone 1	Instantaneous trip of ground distance Zone 1	true
Timer selection for phase and ground distance Zone 2	Timer selection for phase and ground distance Zone 2	Common timer for phase and ground
Timer selection for phase and ground distance Zone 3	Timer selection for phase and ground distance Zone 3	Common timer for phase and ground
Timer selection for phase and ground distance Zone 4	Timer selection for phase and ground distance Zone 4	Common timer for phase and ground

SEL 311L \ Additional Information \ Configuration details \ Trip logic \ Overcurrent

Descrição	Nome	Valor
50P1 Phase overcurrent level 1	50P1 Phase overcurrent level 1	Not used for tripping
50P2 Phase overcurrent level 2	50P2 Phase overcurrent level 2	Definite time
50P3 Phase overcurrent level 3	50P3 Phase overcurrent level 3	Not used for tripping
50G1 Residual overcurrent level 1	50G1 Residual overcurrent level 1	Not used for tripping
50G2 Residual overcurrent level 2	50G2 Residual overcurrent level 2	Definite time Directional
50G3 Residual overcurrent level 3	50G3 Residual overcurrent level 3	Not used for tripping
50G4 Residual overcurrent level 4	50G4 Residual overcurrent level 4	Not used for tripping
50Q1 Negative sequence overcurrent level 1	50Q1 Negative sequence overcurrent level 1	Not used for tripping
50Q2 Negative sequence overcurrent level 2	50Q2 Negative sequence overcurrent level 2	Not used for tripping
50Q3 Negative sequence overcurrent level 3	50Q3 Negative sequence overcurrent level 3	Not used for tripping
50Q4 Negative sequence overcurrent level 4	50Q4 Negative sequence overcurrent level 4	Not used for tripping
51PT Phase time overcurrent	51PT Phase time overcurrent	false
51GT Residual time overcurrent	51GT Residual time overcurrent	false
51QT Negative sequence time overcurrent	51QT Negative sequence time overcurrent	false

SEL 311L \ Additional Information \ Line Current Differential

Descrição	Nome	Valor
Template is Master Template	Template is Master Template	true

SEL 311L \ Additional Information \ Relay Information \ Output contacts

Descrição	Nome	Valor
Contact operating time	Contact operating time	4,00 ms

SEL 311L \ Additional Information \ Relay Information \ Distance Protection - Distance protection

Descrição	Nome	Valor
Rel. Tol Z at Line Angle (SIR<=60)	Rel. Tol Z	5,00 %
Max. Trip time (according operating time curve at SIR=30)	Max. operating time (cycle) - internal operation	2,00 cy
Max. Trip time (according operating time curve at SIR=30)	Max. operating time (ms) - internal operation	33,33 ms
Rel. time tolerance	Rel. t-Tolerance	5,00 %
Pick up / drop off ratio	Pick Up / Drop Off Ratio	0,95
Abs. t-Tolerance	Abs. t-Tolerance- internal operation	1,50 cy

SEL 311L \ Additional Information \ Relay Information \ Distance Protection \ Fault Detectors

Descrição	Nome	Valor
	Rel. I-Tolerance	3,00 %
	Abs. I-Tolerance	0,05 A

SEL 311L \ Additional Information \ Relay Information \ Overcurrent Protection \ Time Overcurrent Elements

Descrição	Nome	Valor
	abs. current tolerance	0,05 A
	rel. current tolerance	3,00 %
	abs. time tolerance	1,75 cy
	rel. time tolerance	0,10 %
	Pickup/Drop off ratio	1,00

SEL 311L \ Additional Information \ Relay Information \ Overcurrent Protection \ Definite Time/Instantaneous Elements

Descrição	Nome	Valor
	abs. current tolerance	0,05 A
	rel. current tolerance	3,00 %
	abs. time tolerance	1,75 cy
	rel. time tolerance	0,10 %
	Pickup/Drop off ratio	1,00
	Max operating time - internal operation	1,50 cy

SEL 311L \ Additional Information \ Relay Information \ Differential Protection

Descrição	Nome	Valor
	87L relative Tolerance	3,00 %
	87L absolute Tolerance	5,00 mA
	Angle Tolerance	3 °
	87L Trip Time	40,00 ms
	87L absolute Time Tolerance	25,00 ms

SEL 311L \ Additional Information \ Circuit Breaker Information

Descrição	Nome	Valor
52ab Percentage	52ab Percentage	20,00 %
CB Trip Time	CB Trip Time	50,00 ms
CB Close Time	CB Close Time	100,00 ms
t-Tolerance abs	t-Tolerance abs	0,00 ms
t-Tolerance rel	t-Tolerance rel	5,00 %

SEL 311L \ Additional Information \ Test Tolerance

Descrição	Nome	Valor
Global Test Tolerance	Test Tolerance	2,00 %

Objeto sob Teste - Outras Funções RIO

CB Configuration

Descrição	Nome	Valor
Tempo de disparo do DJ	Tempo de disparo do DJ	50,00 ms
Tempo de fechamento do DJ	Tempo de fechamento do DJ	100,00 ms
Tempos para 52a, 52b em porcentagem de tempo CB	52a, 52b % de DJ	20,00 %

Equipamento em teste - Ajustes de Distância

Parâmetros do sistema:

Comprimento da linha:	1,230 Ω	Ângulo da linha:	79,39 °
Conexão TP:	na linha	Ponto de Neutro do TC:	Dir.linha
Correção da impedância 1A/I nom:	Não		
Impedância em valores primários:	Não		

Tolerâncias:

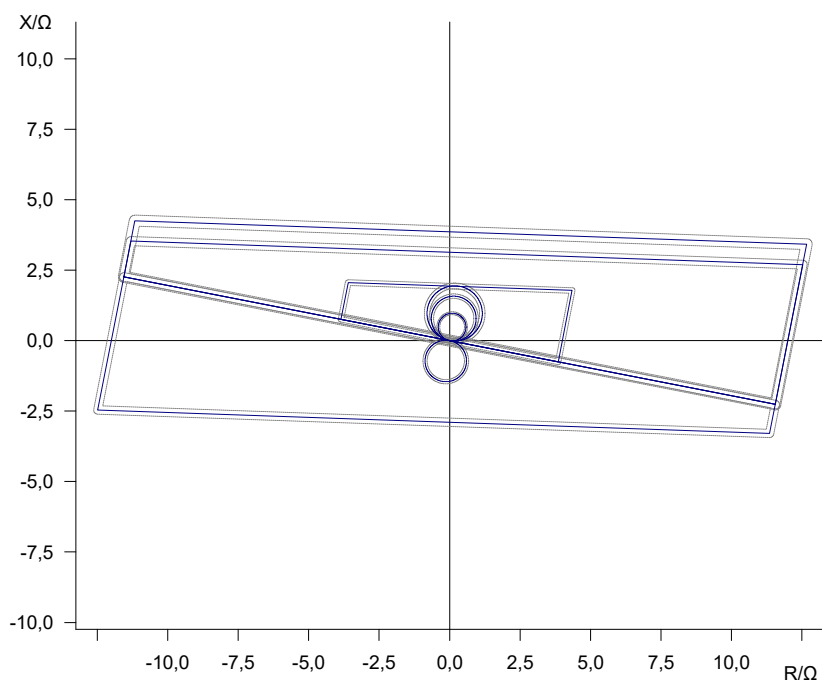
Tol. T rel.:	5,000 %	Tol. T abs. -:	29,00 ms
Tol. T abs. +:	29,00 ms	Tol. Z abs.:	50,00 m Ω
Tol. rel. Z:	5,000 %		

Fator de terra:

Mag. kL:	0,000000	Ângulo kL:	0,000000°
Separar a resistência de arco:	Não		

Ajustes de zona:

Etiqueta	Tipo	Loop de Falta	Tempo de Disparo	Tol.T rel	Tol.T abs+	Tol.T abs-	Tol.Z rel.	Tol.Z abs
ZL1	Disparo	L-L	33,33 ms	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 m Ω
ZL2	Disparo	L-L	433,3 ms	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 m Ω
ZL3	Disparo	L-L	1,533 s	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 m Ω
ZL4	Disparo	L-L	1,533 s	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 m Ω
ZX1	Disparo	L-N	33,33 ms	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 m Ω
ZX2	Disparo	L-N	433,3 ms	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 m Ω
ZX3	Disparo	L-N	1,533 s	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 m Ω
ZX4	Disparo	L-N	1,533 s	5,000 %	29,00 ms	29,00 ms	5,000 %	50,00 m Ω



Referências de ligação XRIO

Nome de referência	Unidade	Valor	Caminho XRIO
RIO.DEVICE.NOMINALVALUES.INOM	In	5,00 A	RIO/Dispositivo/Valores Nominais/In
RIO.DEVICE.NOMINALVALUES.VNOM	V_nom	109,99 V	RIO/Dispositivo/Valores Nominais/V nom

Comentário

Módulo de teste

Nome: OMICRON Advanced Distance Versão: 5.00 SR 2
Início do teste: 29-abr-2026 07:30:53 Fim do Teste: 29-abr-2026 07:31:45
Nome do Usuário: Gerente:
Companhia:

Ajustes do teste

Modelo de teste:

Modelo de teste: Corrente de teste constante ITeste: 10,00 A
Permitir redução de Não kS = kL: Não
ITeste/VTeste:
Mag. ZS: 0,00 Ω Ângulo ZS: 0,00 °
mag. kS: 1,000 Ângulo kS: 0,00 °

Início da Falta:

Modo: Aleatório Ângulo
Offset DC: Não n/d

Tempos:

Pré-falta: 1,000 s Max-falta: 6,000 s
Pós-falta: 500,0 ms Referência de Tempo: Início da falta

Outro:

Zonas extendidas: Inativo Desligar na passagem Sim
Corrente de carga Não por zero:
habilitada: Corrente de carga:: n/d

Ajustes de busca:

Busca res. rel.: 1,000 % Busca res. abs.: 50,00 mΩ
Ignorar característica Não
nominal:
Intervalo de busca: 200,0 mΩ

Entradas Binárias:

Lógica trigger: OU

Nome	Estado de Trigger
Entr. bin. 1	1
Entr. bin. 2	X

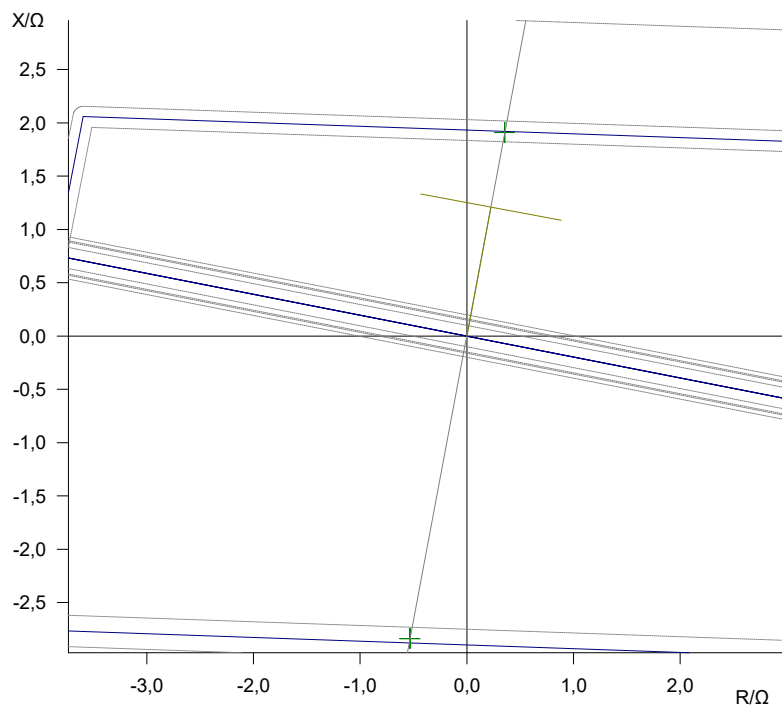
Saídas binárias:

Nome	Tempo de atraso do início da falta	Inclinação	Tempo de atraso de disparo	Inclinação
Saída bin. 1	n/d	Aberto	n/d	Aberto

Resultados do teste

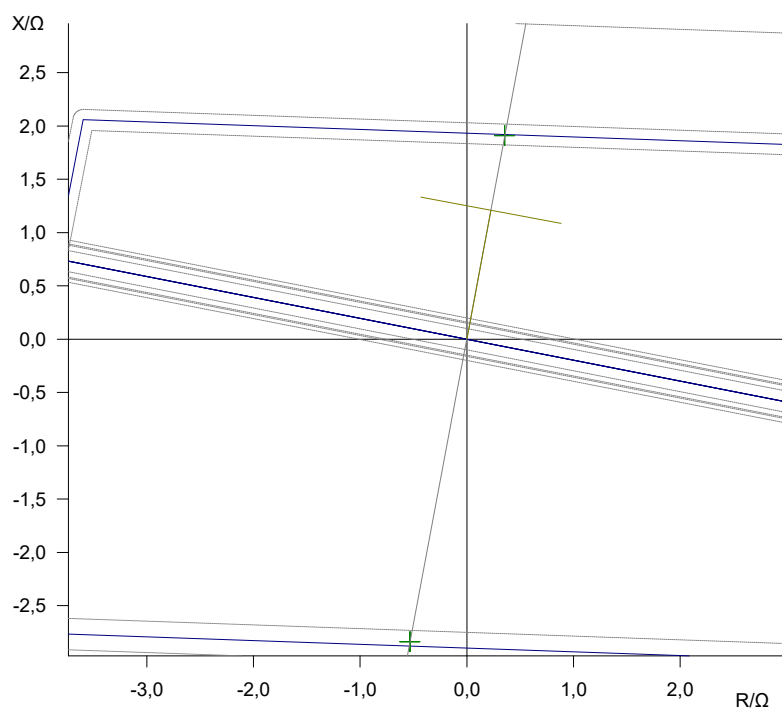
Teste de disparo: Tipo de falta A-N

$ Z $	Phi	%	% de	t nom	t real	Desv.	ITeste	Resultado
1,940 Ω	79,39 °	n/d		33,33 ms	75,20 ms	125,6 %	10,00 A	Aprovado
3,200 Ω	79,39 °	n/d		1,533 s	1,529 s	-0,2761 %	10,00 A	Aprovado
3,887 Ω	79,39 °	n/d		1,533 s	1,529 s	-0,263 %	10,00 A	Aprovado
2,886 Ω	-100,61 °	n/d		1,533 s	1,531 s	-0,1652 %	10,00 A	Aprovado



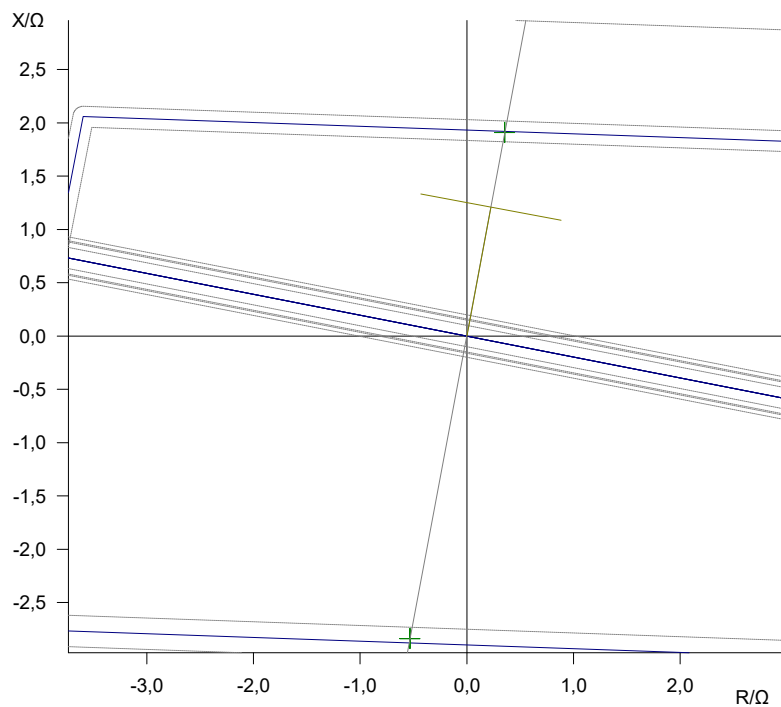
Teste de disparo: Tipo de falta B-N

$ Z $	Phi	%	% de	t nom	t real	Desv.	ITeste	Resultado
1,940 Ω	79,39 °	n/d		33,33 ms	73,90 ms	121,7 %	10,00 A	Aprovado
3,200 Ω	79,39 °	n/d		1,533 s	1,525 s	-0,5174 %	10,00 A	Aprovado
3,887 Ω	79,39 °	n/d		1,533 s	1,527 s	-0,3935 %	10,00 A	Aprovado
2,886 Ω	-100,61 °	n/d		1,533 s	1,531 s	-0,1783 %	10,00 A	Aprovado



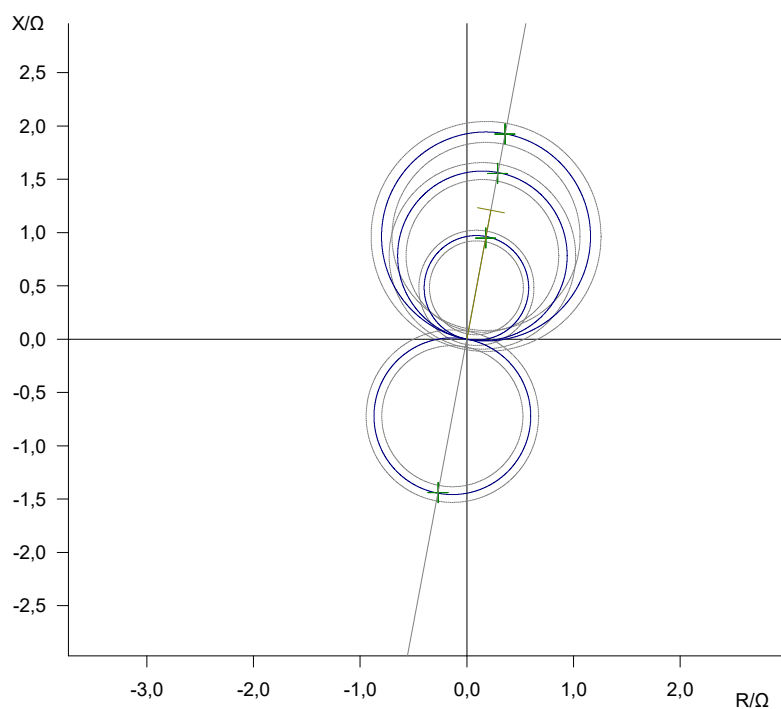
Teste de disparo: Tipo de falta C-N

$ Z $	Phi	%	% de	t nom	t real	Desv.	ITeste	Resultado
1,940 Ω	79,39 °	n/d		33,33 ms	75,40 ms	126,2 %	10,00 A	Aprovado
3,200 Ω	79,39 °	n/d		1,533 s	1,528 s	-0,3739 %	10,00 A	Aprovado
3,887 Ω	79,39 °	n/d		1,533 s	1,530 s	-0,1913 %	10,00 A	Aprovado
2,886 Ω	-100,61 °	n/d		1,533 s	1,530 s	-0,2435 %	10,00 A	Aprovado



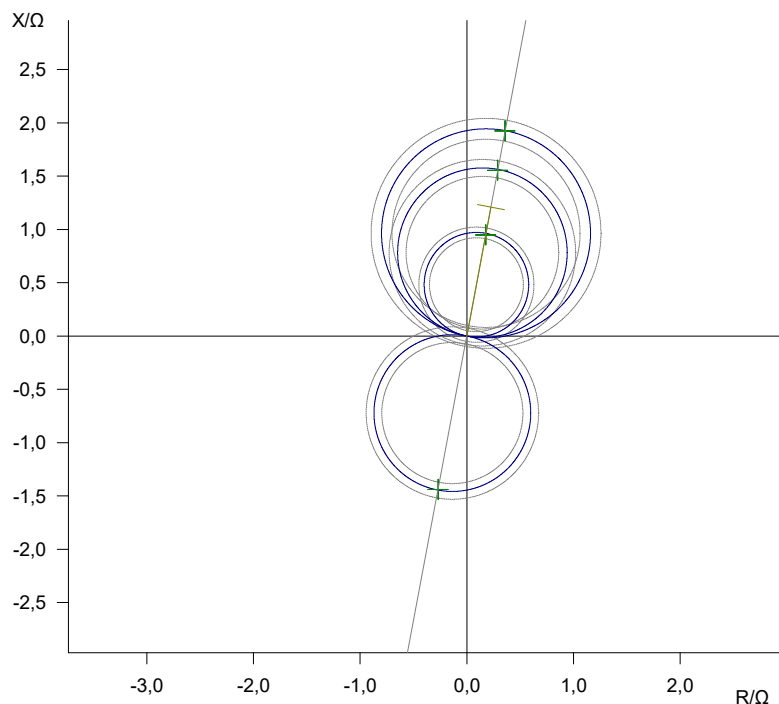
Teste de disparo: Tipo de falta A-B

$ Z $	Phi	%	% de	t nom	t real	Desv.	ITeste	Resultado
964,6 m Ω	79,39 °	n/d		33,33 ms	45,60 ms	36,8 %	10,00 A	Aprovado
1,581 Ω	79,39 °	n/d		433,3 ms	430,1 ms	-0,7462 %	10,00 A	Aprovado
1,956 Ω	79,39 °	n/d		1,533 s	1,528 s	-0,3739 %	10,00 A	Aprovado
1,466 Ω	-100,61 °	n/d		1,533 s	1,529 s	-0,2957 %	10,00 A	Aprovado



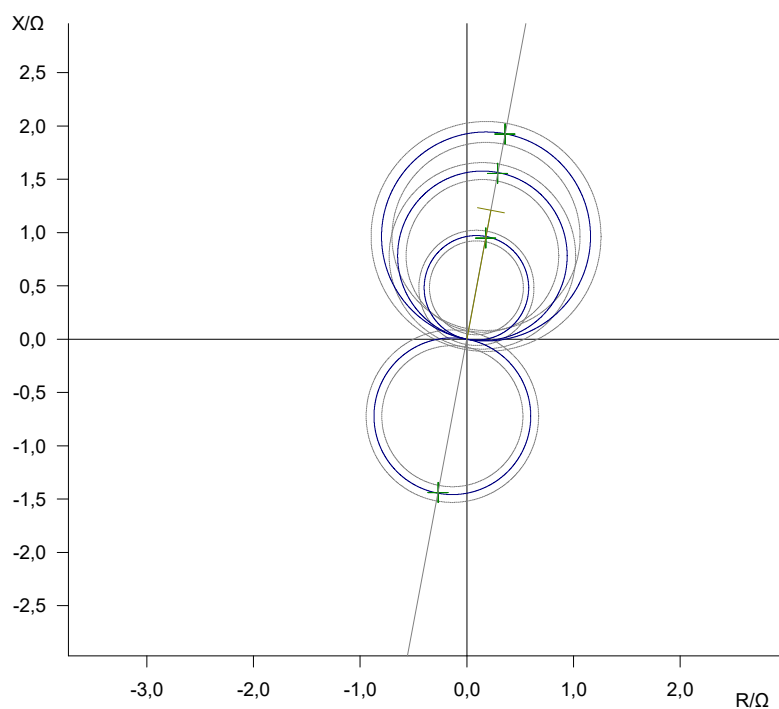
Teste de disparo: Tipo de falta B-C

$ Z $	Phi	%	% de	t nom	t real	Desv.	I _{Teste}	Resultado
964,6 mΩ	79,39 °	n/d		33,33 ms	72,10 ms	116,3 %	10,00 A	Aprovado
1,581 Ω	79,39 °	n/d		433,3 ms	426,9 ms	-1,485 %	10,00 A	Aprovado
1,956 Ω	79,39 °	n/d		1,533 s	1,529 s	-0,3022 %	10,00 A	Aprovado
1,466 Ω	-100,61 °	n/d		1,533 s	1,527 s	-0,413 %	10,00 A	Aprovado



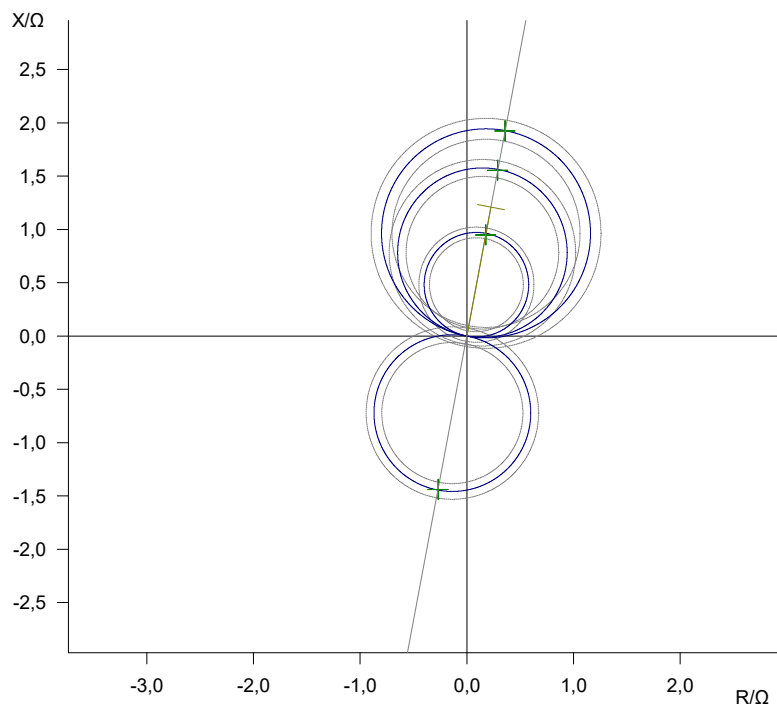
Teste de disparo: Tipo de falta C-A

$ Z $	Phi	%	% de	t nom	t real	Desv.	I _{Teste}	Resultado
964,6 mΩ	79,39 °	n/d		33,33 ms	72,20 ms	116,6 %	10,00 A	Aprovado
1,581 Ω	79,39 °	n/d		433,3 ms	429,7 ms	-0,8385 %	10,00 A	Aprovado
1,956 Ω	79,39 °	n/d		1,533 s	1,527 s	-0,4391 %	10,00 A	Aprovado
1,466 Ω	-100,61 °	n/d		1,533 s	1,530 s	-0,2109 %	10,00 A	Aprovado



Teste de disparo: Tipo de falta A-B-C

Z	Phi	%	% de	t nom	t real	Desv.	ITeste	Resultado
964,6 mΩ	79,39 °	n/d		33,33 ms	69,00 ms	107 %	10,00 A	Aprovado
1,581 Ω	79,39 °	n/d		433,3 ms	427,1 ms	-1,438 %	10,00 A	Aprovado
1,956 Ω	79,39 °	n/d		1,533 s	1,529 s	-0,2696 %	10,00 A	Aprovado
1,466 Ω	-100,61 °	n/d		1,533 s	1,530 s	-0,2435 %	10,00 A	Aprovado



Detalhes do disparo:

Parâmetros:

Tipo de falta:	A-N	Phi:	-100,61 °
Z :	2,886 Ω	X:	-2,837 Ω
R:	-531,4 mΩ	% de:	
%:	n/d		
ITeste:	10,00 A		

Resultados:

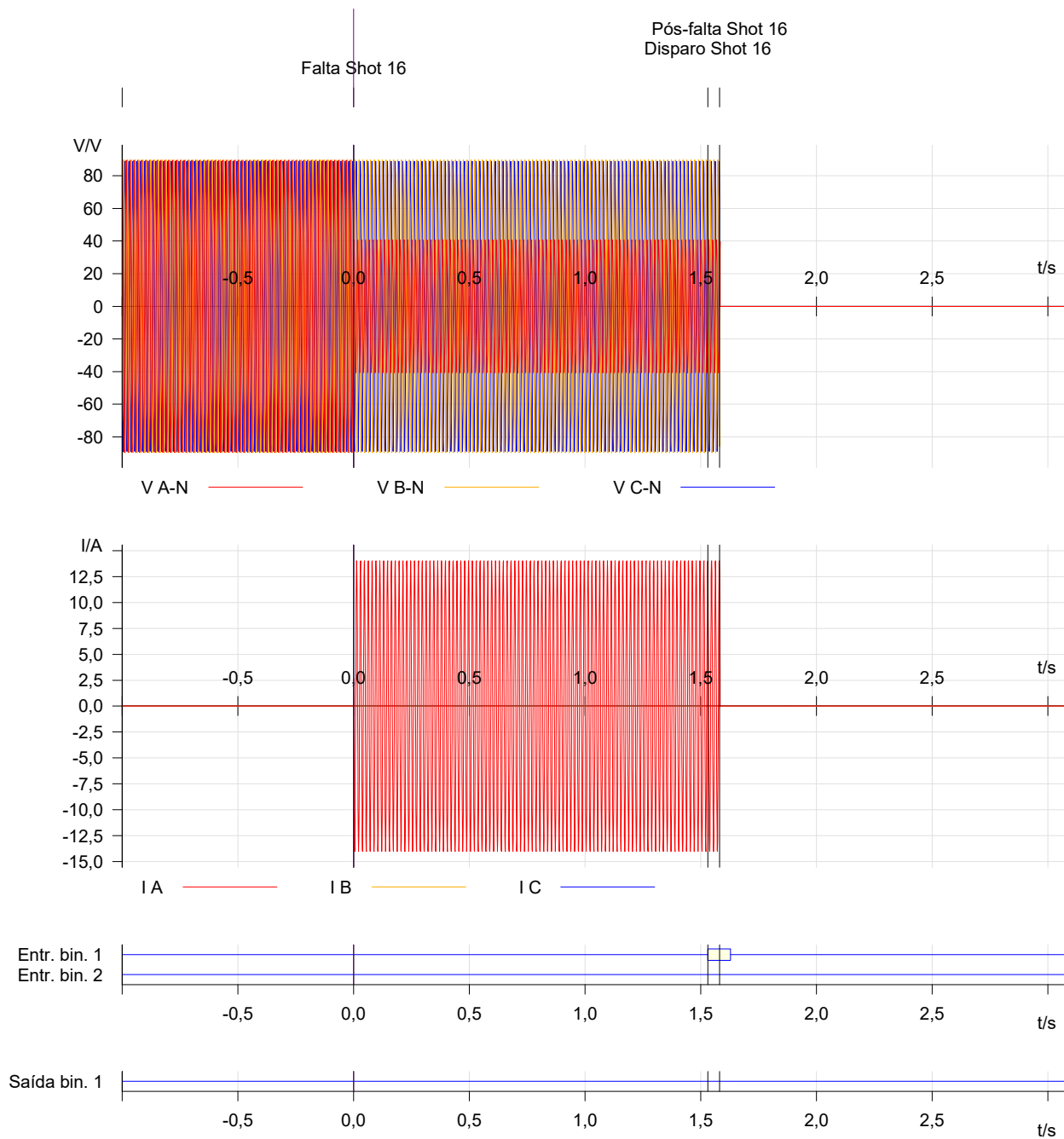
t real:	1,531 s	Avaliação:	Aprovado
t nom:	1,533 s	Desv.:	-0,1652 %
t min:	1,457 s	t max:	sem trip

Quantidades da falta (natural):

VL1:	28,86 V	0,00 °
VL2:	63,50 V	-120,00 °
VL3:	63,50 V	120,00 °
IL1:	10,00 A	100,61 °
IL2:	0,00 A	n/d
IL3:	0,00 A	n/d
VFalta:	28,86 V	0,00 °
IFalta:	10,00 A	100,61 °

Quantidades da falta (simétrico):

V0:	11,55 V	180,00 °
V1:	51,95 V	0,00 °
V2:	11,55 V	180,00 °
I0:	3,333 A	100,61 °
I1:	3,333 A	100,61 °
I2:	3,333 A	100,61 °



Dados do Cursor

	Tempo	Sinal	Valor
Cursor 1	0,00 s	<nenhum>	n/d
Cursor 2	6,000 s	<nenhum>	n/d
C2 - C1	6,000 s		n/d

Estado do teste:
Teste aprovado

Overcurrent:
Equipamento em teste - Parâmetros de Sobrecorrente

Valores - Gerais:

TolAbsdeTempo: 0,03 s
TolReldeTempo: 0,10 %

TolAbsdeCorrente: 0,05 Iref
TolReldeCorrente: 3,00 %
Direcional: Sim

Conexão de TP: No objeto protegido
Conexão de ponto neutro do TP: Para objeto protegido

Elementos - Fase:

Ativo	Nome	Característica de Disparo	I Pick-up	Tempo	Relação de Reset	Direção
Não	51P	U3 Very Inverse	0,10 Iref	2,00	1,00	Não Direcional
Não	50/67P1	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Sim	50/67P2	Definite Time	5,00 Iref	1,53 s	1,00	Não Direcional
Não	50/67P3	Definite Time	28,00 Iref	0,03 s	1,00	Não Direcional

Elementos - Residual:

Ativo	Nome	Característica de Disparo	I Pick-up	Tempo	Relação de Reset	Direção
Não	51G	U3 Very Inverse	0,10 Iref	2,00	1,00	Para frente
Não	50/67G1	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Sim	50/67G2	Definite Time	2,50 Iref	1,53 s	1,00	Para frente
Não	50/67G3	Definite Time	2,50 Iref	0,03 s	1,00	Não Direcional
Não	50/67G4	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional

Elementos - Sequência Negativa:

Ativo	Nome	Característica de Disparo	I Pick-up	Tempo	Relação de Reset	Direção
Não	51Q	U3 Very Inverse	0,10 Iref	2,00	1,00	Não Direcional
Não	50/67Q1	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Não	50/67Q2	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Não	50/67Q3	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional
Não	50/67Q4	Definite Time	0,10 Iref	0,03 s	1,00	Não Direcional

Módulo de teste

Nome: OMICRON Overcurrent
Início do teste: 29-abr-2026 07:36:11
Nome do Usuário:
Companhia:

Versão: 5.00 SR 2
Fim do Teste: 29-abr-2026 07:37:35
Gerente:

Ajustes do teste:

Modelo de Falta:

Referência de Tempo: Início da falta
Corrente de carga: 0,00 A
Ângulo de carga: n/d
Tempo de pré-falta: 100,0 ms
Tempo máximo abs.: 240,0 s
Tempo de pós-falta: 500,0 ms
Tempo máximo rel.: 100,0 %
Habilitar a saída de tensão: Sim
Tensão de falta LN (todas menos faltas bifásicas): 30,00 V
Tensão de falta LL (para faltas bifásicas): 51,96 V
DC decadente ativa: Não
Tempo constante: n/d
Tempo mín. de carac. CB: 50,00 ms
Reinício térmico ativo: Não
Método de reinício térmico: n/a
Mensagem de reinício térmico: n/d

Teste de disparo:

Tipo	Relativo A	Fator	Magnitude	Ângulo	tnom	tmin	tmax
A-B	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
A-B	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,496 s	1,554 s
A-B	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
B-C	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
B-C	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,496 s	1,554 s
B-C	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
C-A	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
C-A	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,496 s	1,554 s
C-A	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
A-B-C	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
A-B-C	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,496 s	1,554 s
A-B-C	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
A	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
A	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,496 s	1,554 s
A	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
B	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
B	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,496 s	1,554 s
B	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
C	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
C	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,496 s	1,554 s
C	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,496 s	1,554 s
A-N	50/67G2	1,100	2,750 A	-60,00 °	1,525 s	1,496 s	1,554 s
A-N	50/67G2	1,300	3,250 A	-60,00 °	1,525 s	1,496 s	1,554 s
A-N	50/67G2	1,500	3,750 A	-60,00 °	1,525 s	1,496 s	1,554 s
B-N	50/67G2	1,100	2,750 A	-60,00 °	1,525 s	1,496 s	1,554 s
B-N	50/67G2	1,300	3,250 A	-60,00 °	1,525 s	1,496 s	1,554 s
B-N	50/67G2	1,500	3,750 A	-60,00 °	1,525 s	1,496 s	1,554 s
C-N	50/67G2	1,100	2,750 A	-60,00 °	1,525 s	1,496 s	1,554 s
C-N	50/67G2	1,300	3,250 A	-60,00 °	1,525 s	1,496 s	1,554 s
C-N	50/67G2	1,500	3,750 A	-60,00 °	1,525 s	1,496 s	1,554 s
3I0	50/67G2	1,100	2,750 A	-60,00 °	1,525 s	1,496 s	1,554 s
3I0	50/67G2	1,300	3,250 A	-60,00 °	1,525 s	1,496 s	1,554 s
3I0	50/67G2	1,500	3,750 A	-60,00 °	1,525 s	1,496 s	1,554 s

Saídas binárias:

Nome	Estado
Saída bin. 1	0
Saída bin. 2	0
Saída bin. 3	0
Saída bin. 4	0

Entradas Binárias:

Lógica trigger: E

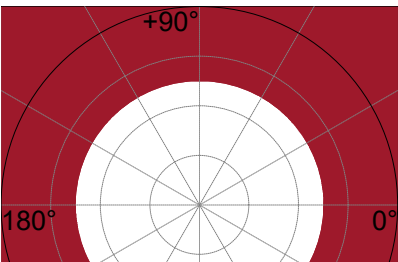
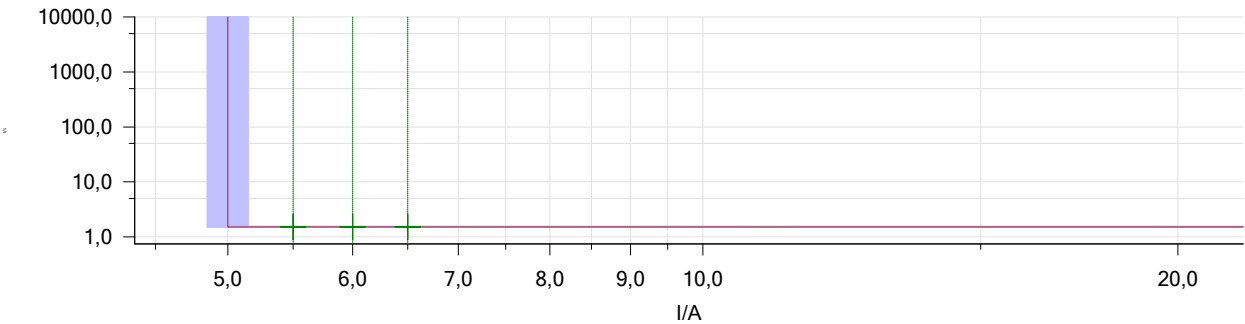
Nome	Estado de Trigger
Entr. bin. 1	1
Entr. bin. 2	X

Resultados de Teste Curtos:

Tipo	Relativo A	Fator	Magnitude	Ângulo	tnom	tatual	Desvio	Sobrecarga	Result
A-B	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,526 s	0,09180 %	Não	Aprova
A-B	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,526 s	0,04590 %	Não	Aprova
A-B	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,527 s	0,1049 %	Não	Aprova
B-C	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,527 s	0,1377 %	Não	Aprova
B-C	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,525 s	-0,006600 %	Não	Aprova
B-C	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,520 s	-0,3541 %	Não	Aprova
C-A	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,526 s	0,05900 %	Não	Aprova
C-A	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,527 s	0,1311 %	Não	Aprova
C-A	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,526 s	0,07210 %	Não	Aprova
A-B-C	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,523 s	-0,1049 %	Não	Aprova
A-B-C	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,521 s	-0,2754 %	Não	Aprova
A-B-C	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,519 s	-0,3803 %	Não	Aprova
A	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,525 s	0,02620 %	Não	Aprova
A	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,525 s	0,00 %	Não	Aprova
A	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,525 s	-0,01310 %	Não	Aprova
B	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,528 s	0,2098 %	Não	Aprova
B	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,527 s	0,1311 %	Não	Aprova
B	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,524 s	-0,03930 %	Não	Aprova
C	50/67P2	1,100	5,500 A	-60,00 °	1,525 s	1,524 s	-0,04590 %	Não	Aprova
C	50/67P2	1,200	6,000 A	-60,00 °	1,525 s	1,524 s	-0,03930 %	Não	Aprova
C	50/67P2	1,300	6,500 A	-60,00 °	1,525 s	1,527 s	0,1049 %	Não	Aprova
A-N	50/67G2	1,100	2,750 A	-60,00 °	1,525 s	1,526 s	0,03280 %	Não	Aprova
A-N	50/67G2	1,300	3,250 A	-60,00 °	1,525 s	1,526 s	0,08520 %	Não	Aprova
A-N	50/67G2	1,500	3,750 A	-60,00 °	1,525 s	1,521 s	-0,2492 %	Não	Aprova
B-N	50/67G2	1,100	2,750 A	-60,00 °	1,525 s	1,527 s	0,1508 %	Não	Aprova
B-N	50/67G2	1,300	3,250 A	-60,00 °	1,525 s	1,520 s	-0,3541 %	Não	Aprova
B-N	50/67G2	1,500	3,750 A	-60,00 °	1,525 s	1,518 s	-0,4787 %	Não	Aprova
C-N	50/67G2	1,100	2,750 A	-60,00 °	1,525 s	1,528 s	0,2164 %	Não	Aprova
C-N	50/67G2	1,300	3,250 A	-60,00 °	1,525 s	1,527 s	0,09840 %	Não	Aprova
C-N	50/67G2	1,500	3,750 A	-60,00 °	1,525 s	1,522 s	-0,2164 %	Não	Aprova
3I0	50/67G2	1,100	2,750 A	-60,00 °	1,525 s	1,527 s	0,1574 %	Não	Aprova
3I0	50/67G2	1,300	3,250 A	-60,00 °	1,525 s	1,525 s	-0,02620 %	Não	Aprova
3I0	50/67G2	1,500	3,750 A	-60,00 °	1,525 s	1,520 s	-0,3016 %	Não	Aprova

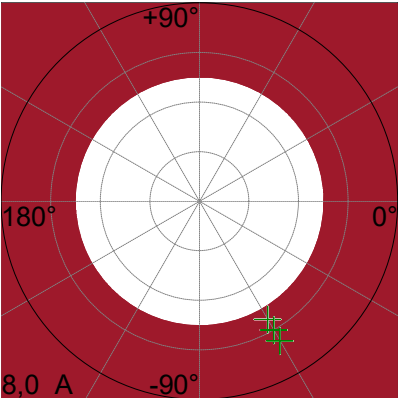
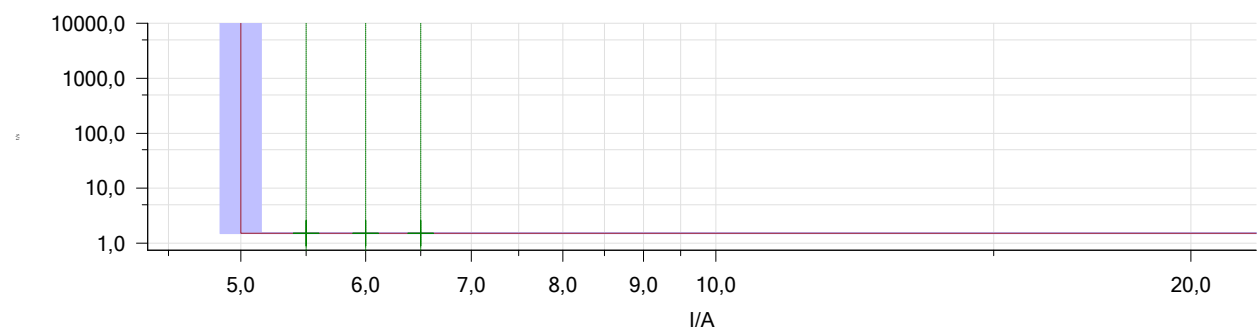
Gráficos para Tipos de Falta:

Tipo	Ângulo
A-B	-60,00 °



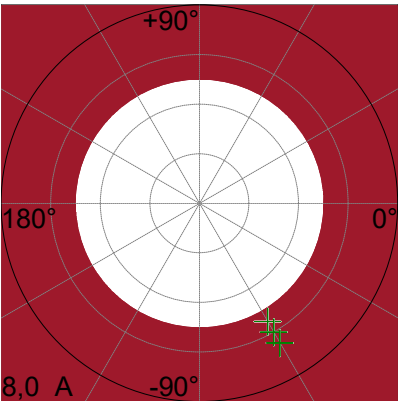
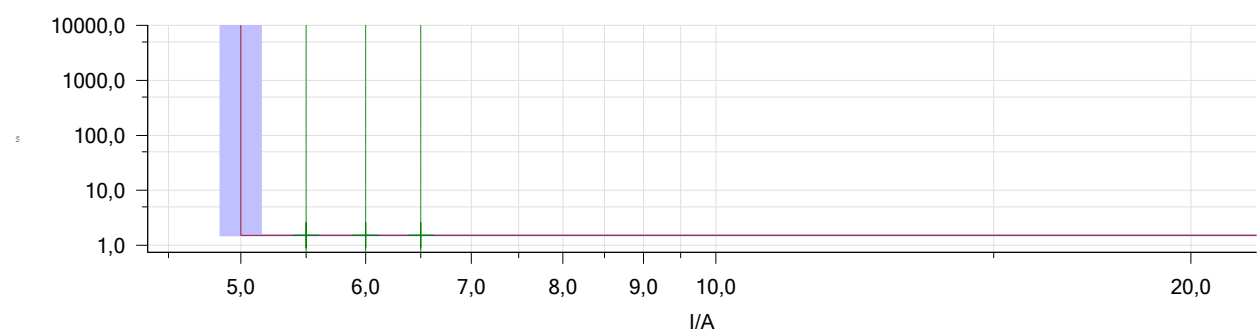
Gráficos para Tipos de Falta:

Tipo	Ângulo
B-C	-60,00 °



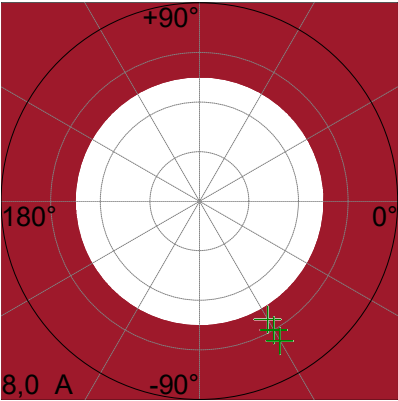
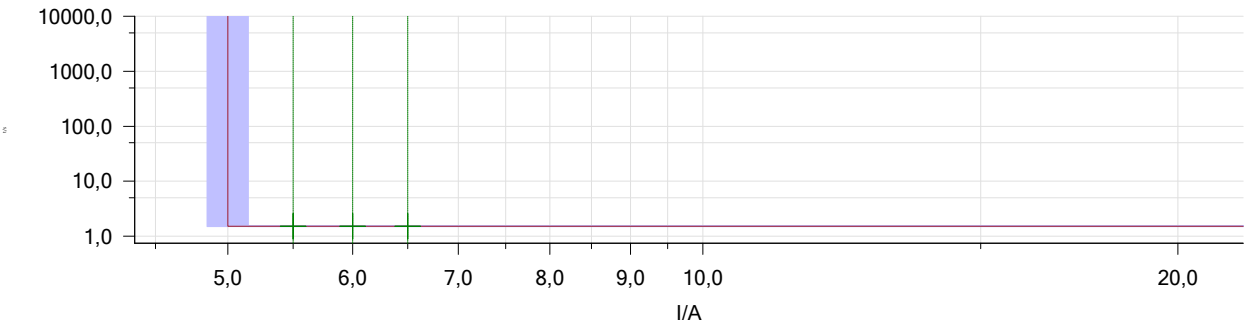
Gráficos para Tipos de Falta:

Tipo	Ângulo
C-A	-60,00 °



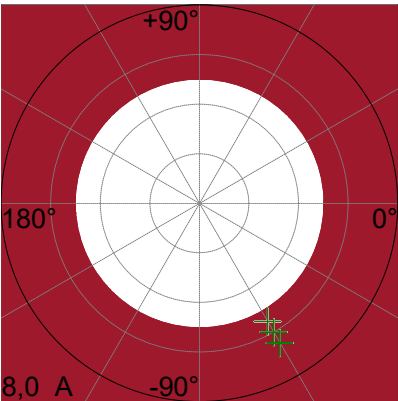
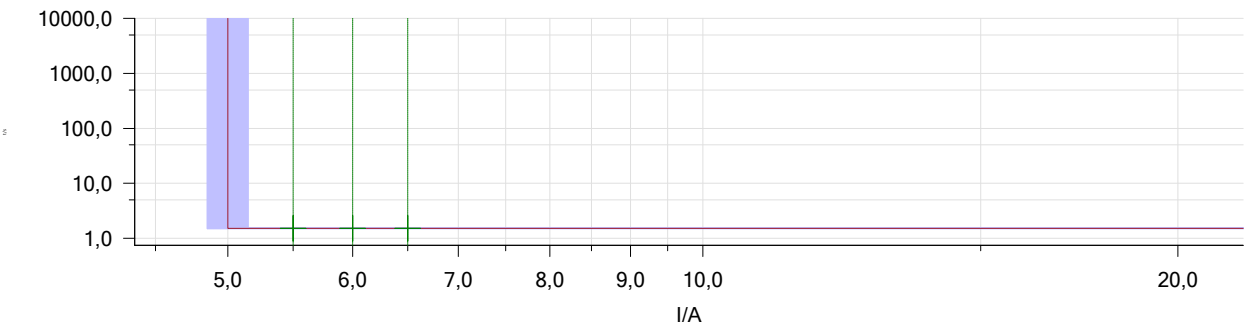
Gráficos para Tipos de Falta:

Tipo	Ângulo
A-B-C	-60,00 °



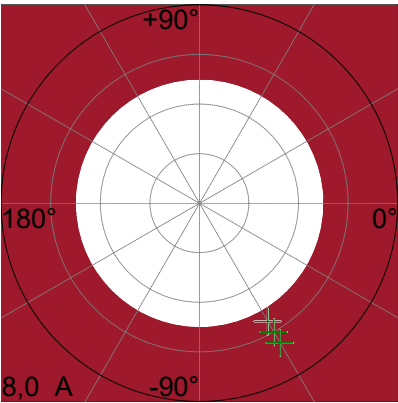
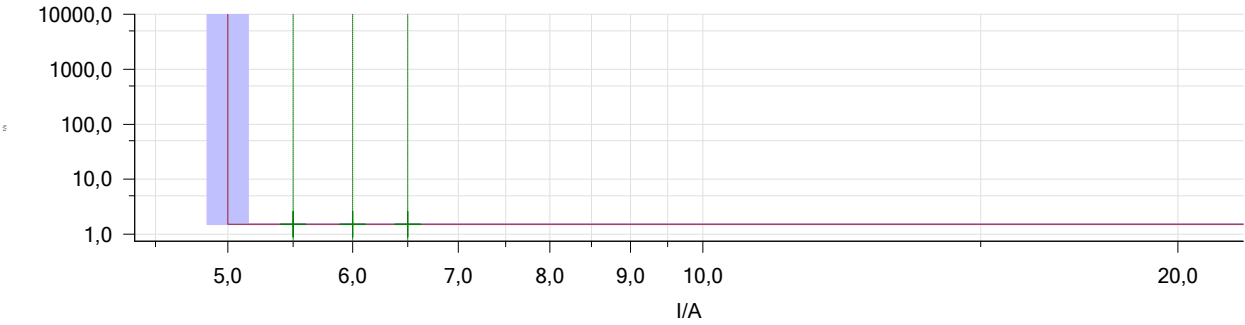
Gráficos para Tipos de Falta:

Tipo	Ângulo
A	-60,00 °



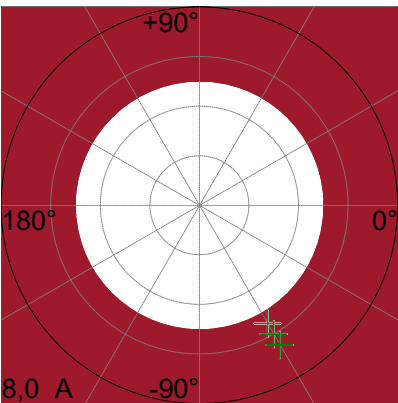
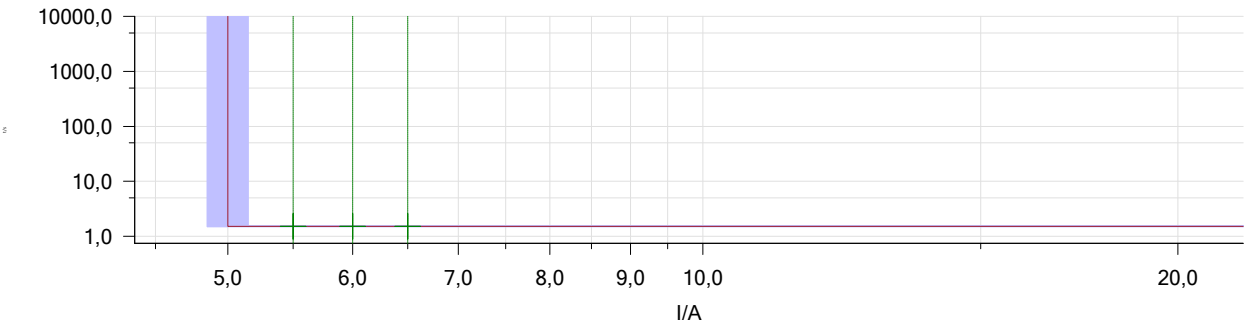
Gráficos para Tipos de Falta:

Tipo	Ângulo
B	-60,00 °



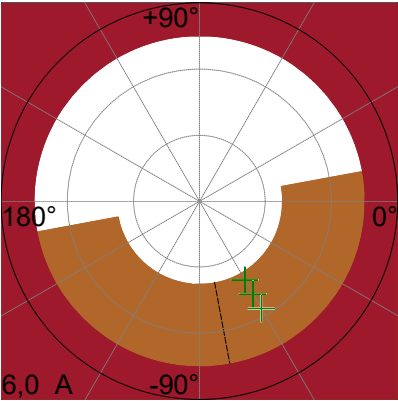
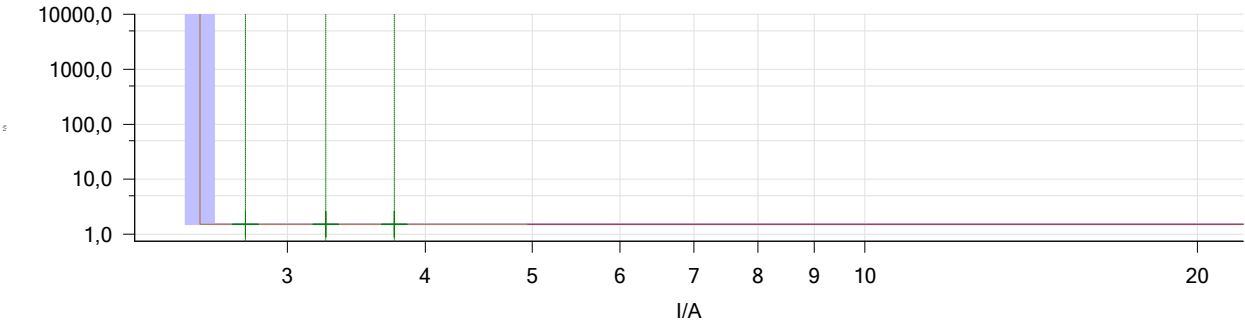
Gráficos para Tipos de Falta:

Tipo	Ângulo
C	-60,00 °



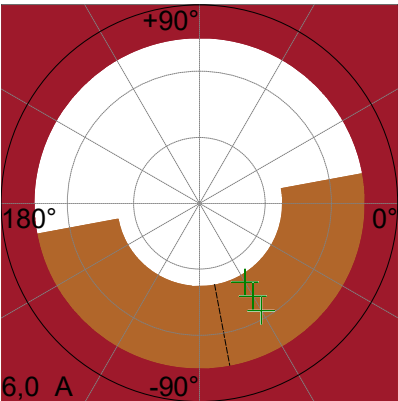
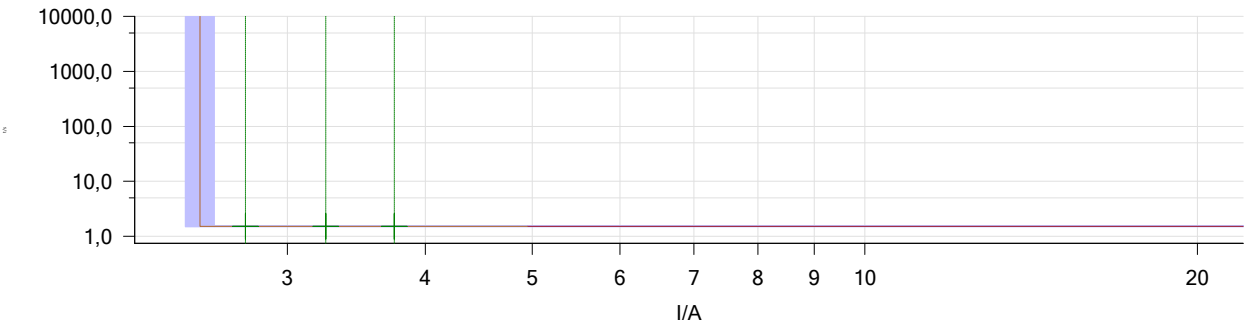
Gráficos para Tipos de Falta:

Tipo	Ângulo
A-N	-60,00 °



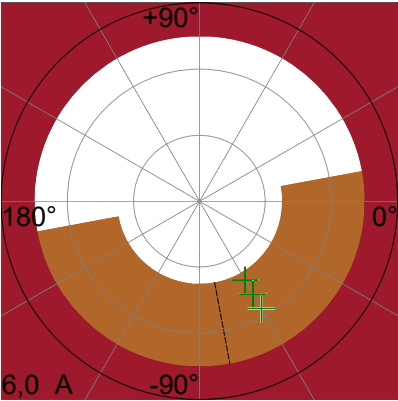
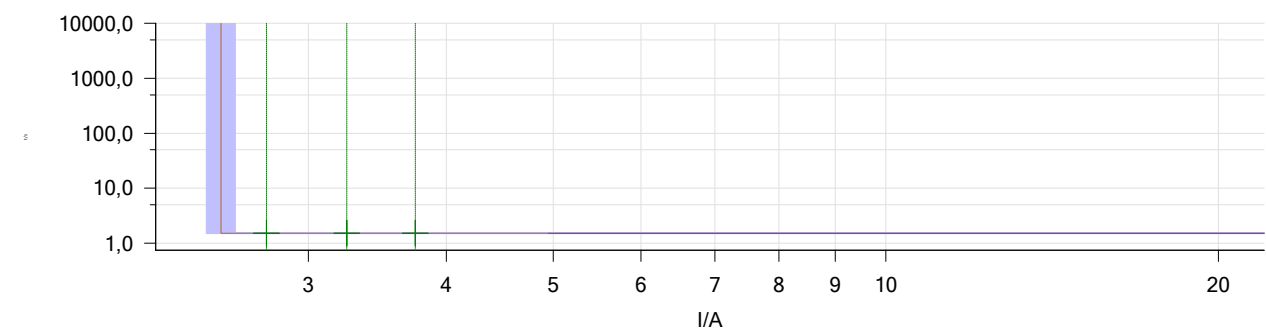
Gráficos para Tipos de Falta:

Tipo	Ângulo
B-N	-60,00 °



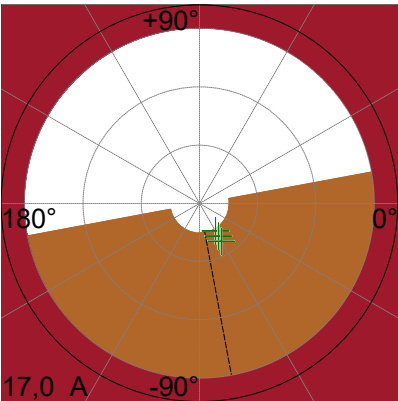
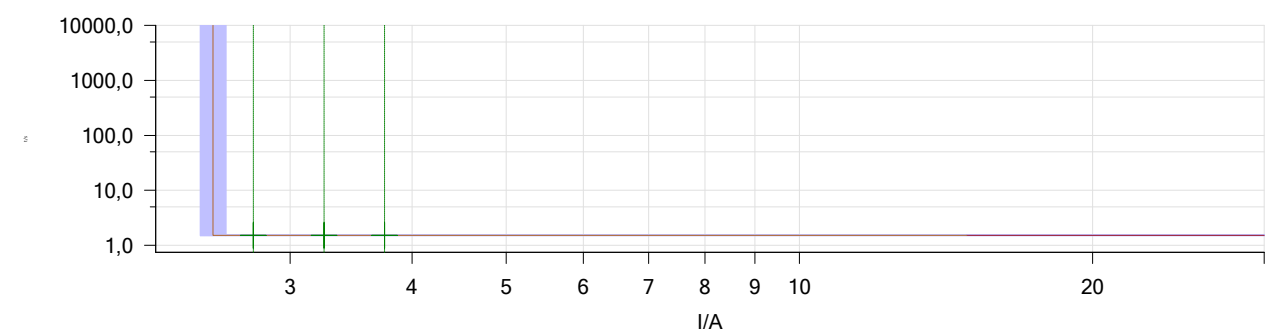
Gráficos para Tipos de Falta:

Tipo	Ângulo
C-N	-60,00 °



Gráficos para Tipos de Falta:

Tipo	Ângulo
3I0	-60,00 °



Estado do teste:

33 de 33 pontos testados.

33 pontos aprovados.

0 pontos reprovados.

Teste aprovado